Safety in Embedded Control Systems

Hardware

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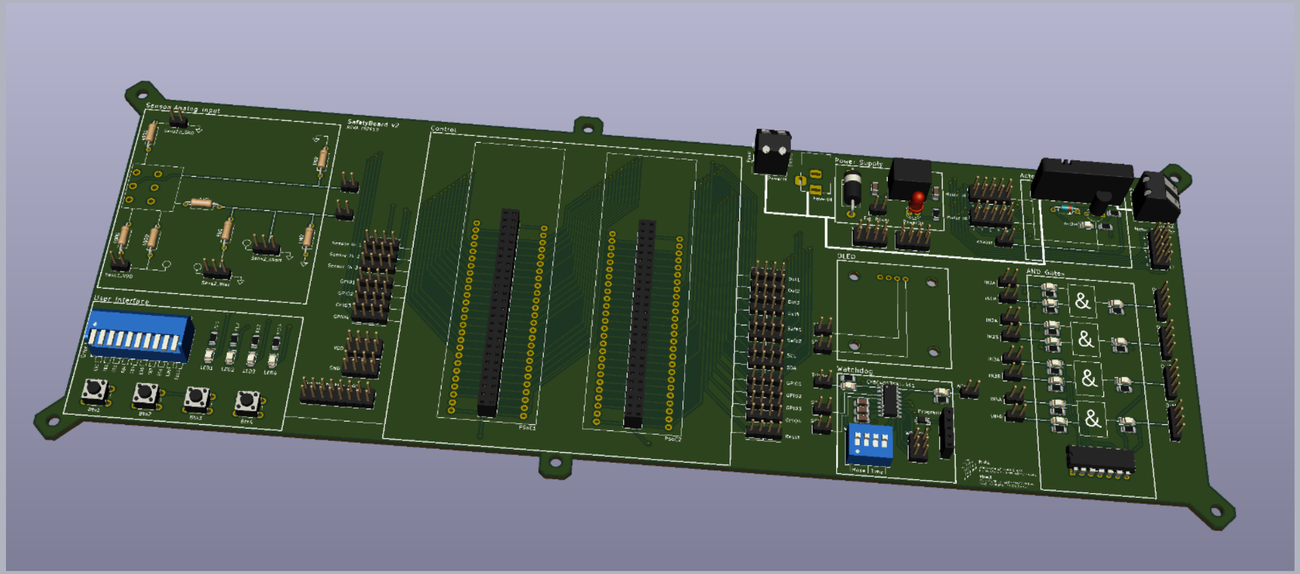


Content

- From functional to hardware architecture
- Hardware component reliability
- Diagnostic Coverage
- Common Cause Analysis
- From component reliability to system reliability
- Safety microcontroller
- The mystery of pins

5.2 MICROCIRCUITS, MEMORIES

DESCRIPTION
1. Read Only Memories (ROM)
2. Programmable Read Only Memories (PROM)
3. Ultraviolet Eraseable PROMs (UVEPROM)
4. "Flash," MNOS and Floating Gate Electrically Eraseable PROMs (EEPROM). Includes both floating gate tunnel oxide (FLOTOX) and textured polysilicon type EEPROMs
5. Static Random Access Memories (SRAM)
6. Dynamic Random Access Memories (DRAM)

$\lambda_p = (C_1 \pi_T + C_2 \pi_E + \lambda_{cyc}) \pi_Q \pi_L$ Failures/10^6 Hours

Die Complexity Failure Rate - C_1

	MOS				Bipolar	
Memory Size, B (Bits)	ROM	PROM, UVEPROM, EEPROM, EAPROM	DRAM	SRAM (MOS & BiMOS)	ROM, PROM	SRAM
Up to 16K	.00065	.00085	.0013	.0078	.0094	.0052
16K < B ≤ 64K	.0013	.0017	.0025	.016	.019	.011
64K < B ≤ 256K	.0026	.0034	.0050	.031	.038	.021
256K < B ≤ 1M	.0052	.0068	.010	.062	.075	.042

A_1 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Flotox[1]	Textured-Poly[2]

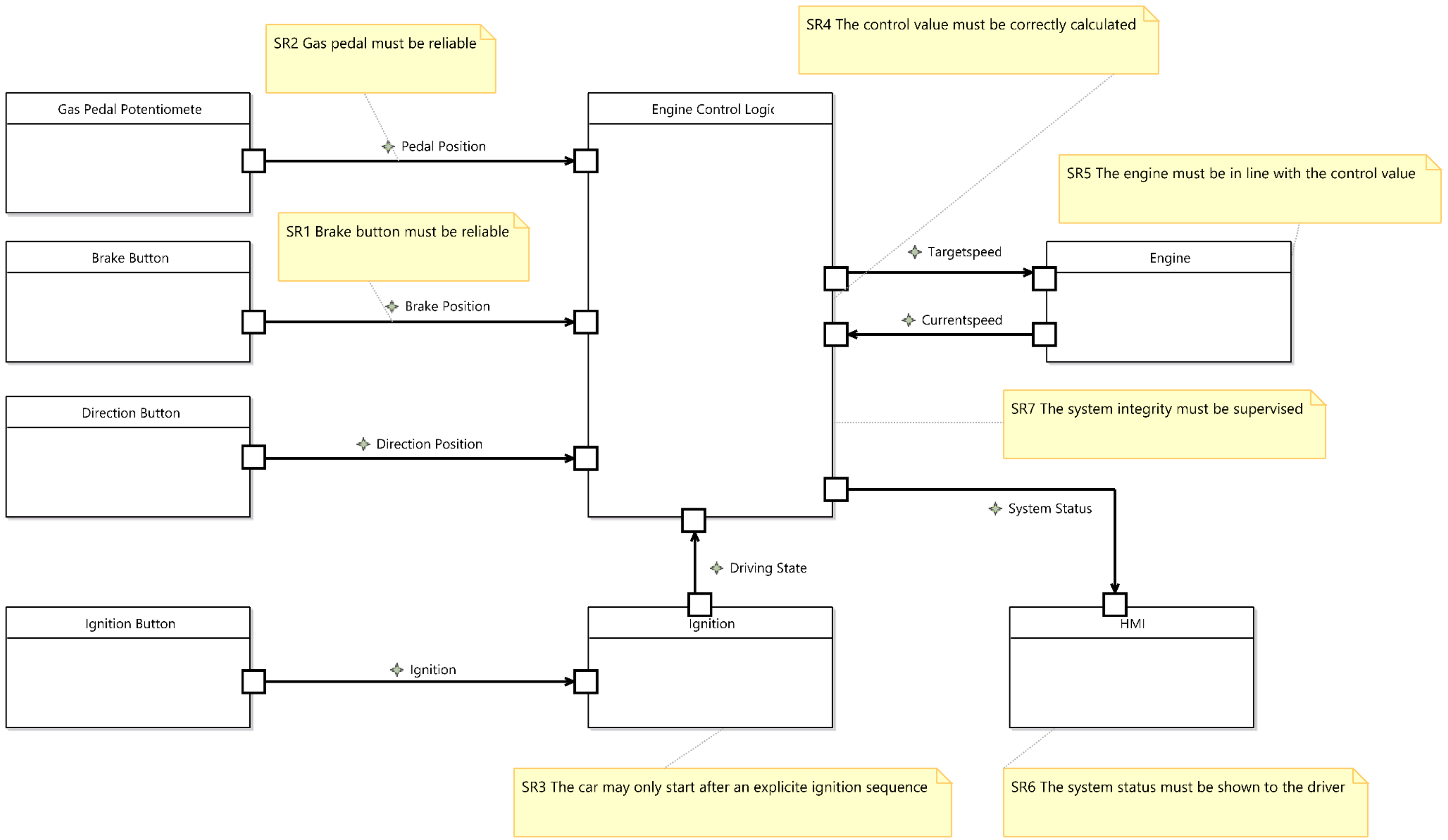
A_2 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Textured-Poly A_2
Up to 300K	0

Result of the Hazard and Risk Analysis: A set of Safety Requirements and the required ASIL System Level

Id	Requirements	$ASIL_{req}$
SR1	The brake signal must be reliable	C
SR2	The gas pedal signal must be reliable	C
SR3	The car may only start after an explicit ignition sequence	QM
SR4	The control speed value must be correctly calculated	C
SR5	The engine currentspeed must be in line with the controlspeed value	C
SR6	The system status must be shown to the driver	A
SR7	The system integrity must be supervised	C

A first functional architecture



Safety Mechanisms

Component Reliability

- MTTF
- Process

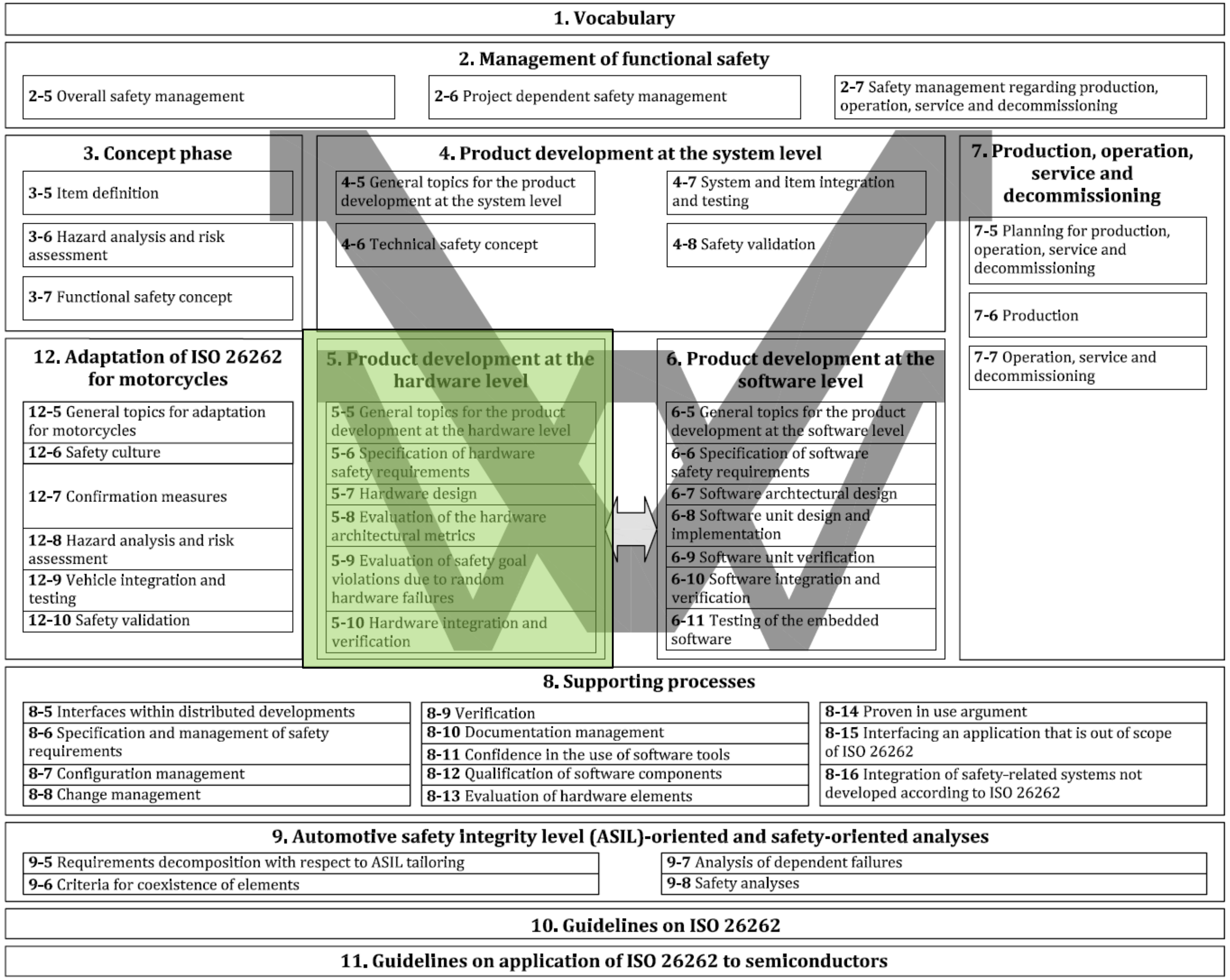
Fault Detection

- Diagnostic Coverage
- Error Handling

Architecture

- Redundancy
- Freedom from Interferance
- Common Cause Failure
- Safe State

ISO26262 Part 5 – Product development at the hardware level



The necessary activities and processes for the product development at the hardware level include:

— the hardware implementation of the technical safety concept;

— the analysis of potential hardware faults and their effects; and

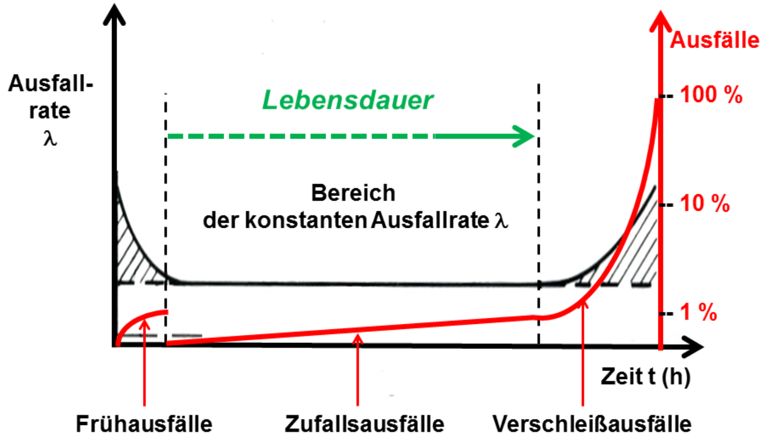
— the coordination with software development.

Characteristics of Hardware Failures

- Hardware failures typically are random failures
- The Failure rate λ(t) describes the number of failures over time, 1 FIT = 1 x 10^{-9}/h
- The survival propability of a component is assumed to follow the Weibull curve

$$R(t) = e^{-(\frac{t}{T})^k} \quad \text{with} \quad \begin{array}{l} k < 1 \\ k = 1 \\ k > 1 \end{array} \quad \begin{array}{l} \Rightarrow \text{Early Failures (often design related)} \\ \Rightarrow \text{Random Failure} \\ \Rightarrow \text{Wear off Failure} \end{array}$$

T: Lifetime,
63,2% broken



Hardware Safety Requirements

The hardware safety requirements specification shall include each hardware requirement that relates to functional safety, including the following:

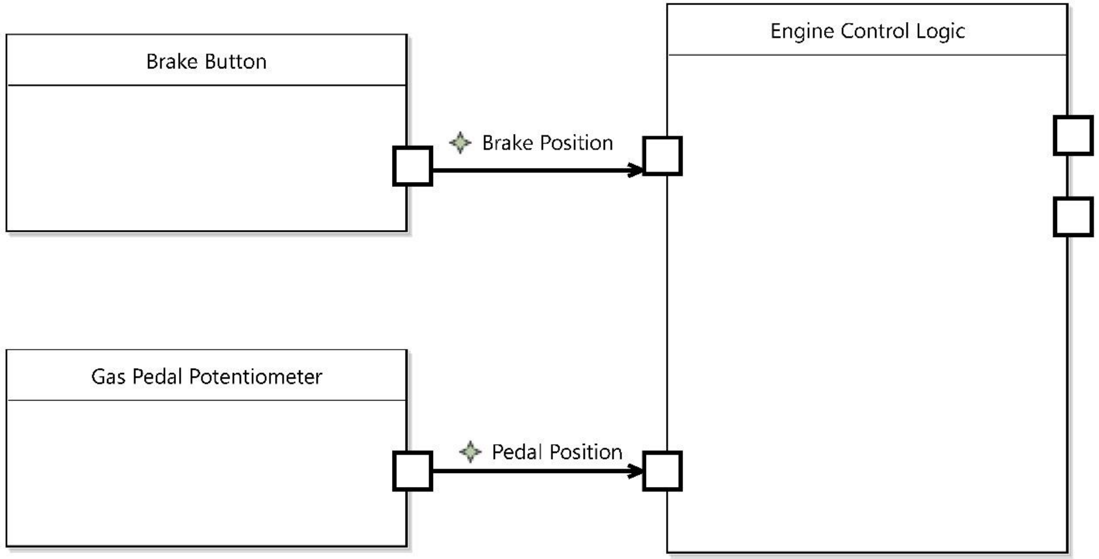
- the hardware safety requirements and relevant properties of safety mechanisms to control **internal failures** of the hardware of the element. These include internal safety mechanisms to cover transient faults when shown to be relevant due to, for instance, the technology used; (e.g. watchdogs for an MCU)
- the hardware safety requirements and relevant properties of safety mechanisms to control or **tolerate failures external** to the element; (e.g. open circuits)
- the hardware safety requirements and relevant properties of safety mechanisms to **comply with the safety requirements of other elements**; (e.g. external sensors or actuators)
- the hardware safety requirements and relevant properties of safety mechanisms to **detect and signal internal or external failures**; (e.g. fault tolerant time interval)
- the hardware safety requirements **not specifying safety mechanisms**. (e.g. robustness requirements)

Hardware Design Part 1 – Sensor Side

- SR 1 & SR 2 - The brake / gas pedal signal must be reliable (ASIL C)

- The sensor hardware must be reliable
- The transmission must be reliable
- The port pin must be reliable
- The peripheral must be reliable

- And of course the software, but that's another story



Hardware safety analysis and requirements

	Control Internal Failure	Tolerate External Failure	Safety requirements compliance	Failure Detection	Additional hardware requirements
Potentiometer	• Wear off sliding contact	• Open or short circuit of any contact		• Range check • Signal comparison • Redundancy	
Analog transmission		• Cross talk		• Inverted signals • Redundancy	
Pin	• Configuration • Sticky pin • Power on behavior	• Open or short circuit		• Selftest • Redundancy	
Peripheral	• Configuration			• Selftest • Redundancy	

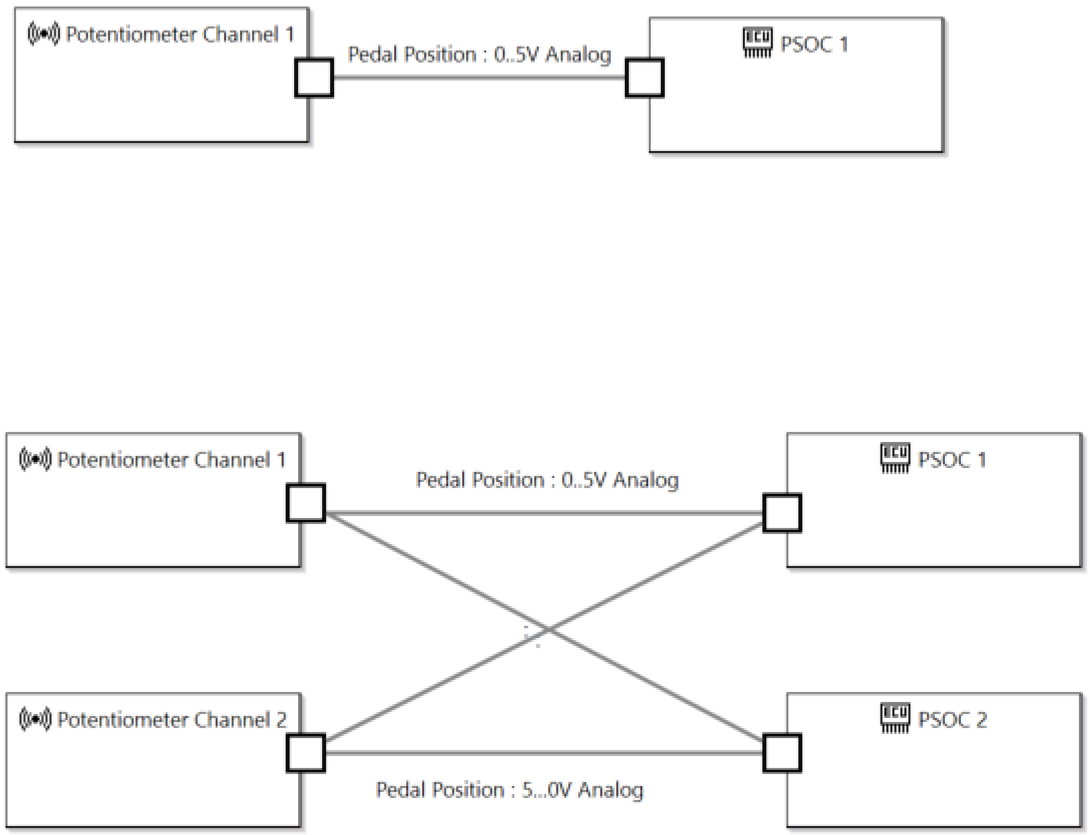
ISO26262 methods for hardware architecture

Table 1 — Properties of hardware architectural design

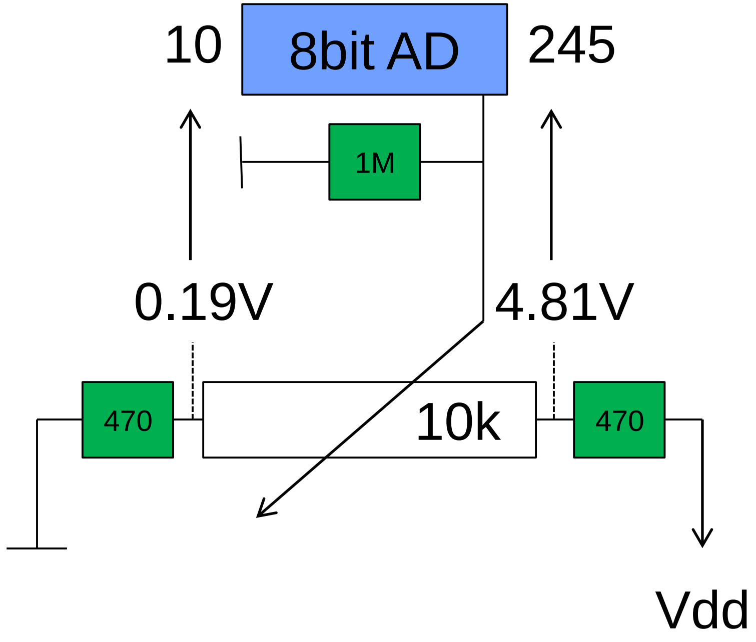
Properties		ASIL A	ASIL B	ASIL C	ASIL D
1	Hierarchical design	+	+	+	+
2	Precisely defined interfaces of safety-related hardware components	++	++	++	++
3	Avoidance of unnecessary complexity of interfaces	+	+	+	+
4	Avoidance of unnecessary complexity of hardware components	+	+	+	+
5	Maintainability (service)	+	+	++	++
6	Testability[a]	+	+	++	++

[a] Testability includes testability during development, production, service and operation.

A first hardware design (focus pedal sensor only)



Range: 0..255
Valid Range: 11..244
Error Range: 0..10 and 245..255



How to deal with hardware faults

Clause 8 describes two metrics to evaluate the effectiveness of the hardware architecture of the item and the implemented safety mechanisms to cope with random hardware failures.

As a complement to Clause 8, Clause 9 describes two alternative methods to evaluate whether the residual risk of safety goal violations is sufficiently low, either by using a global probabilistic approach (see 9.4.2, PMHF method) or by using a cut-set analysis (see 9.4.3, EEC method) to study the impact of each identified fault of a hardware element upon the violation of the safety goals.

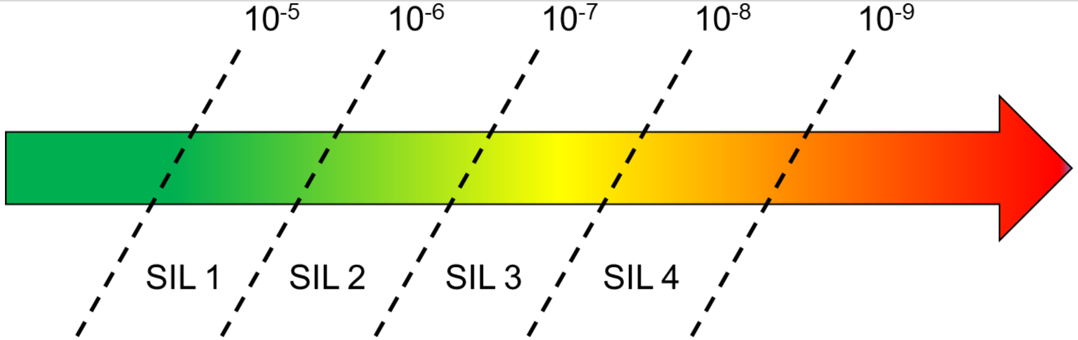
Is the safety architecture capable to deal with random hardware faults

Do we have inter-dependabilities, which must be considered.

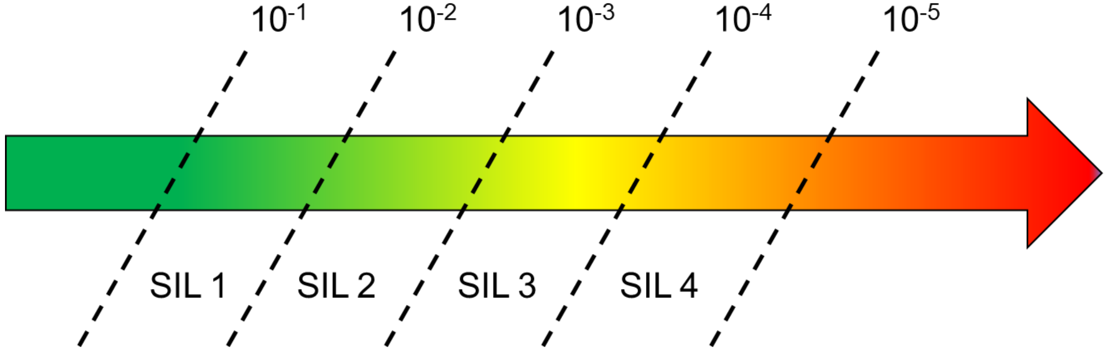
Hardware Failure Rate and SIL/ASIL
Probabilistic Metric for Random Hardware Failures

IEC61508 SIL

- Safe System Function
- Probability of Failure per Hour



- Protection Function
- Activation less than once per year
- Probability of Failure per Demand



ISO26262 ASIL

- Probability of Failure per Hour

Note: ASIL A does not require certified HW

Table 6 — Possible source for the derivation of the random hardware failure target values

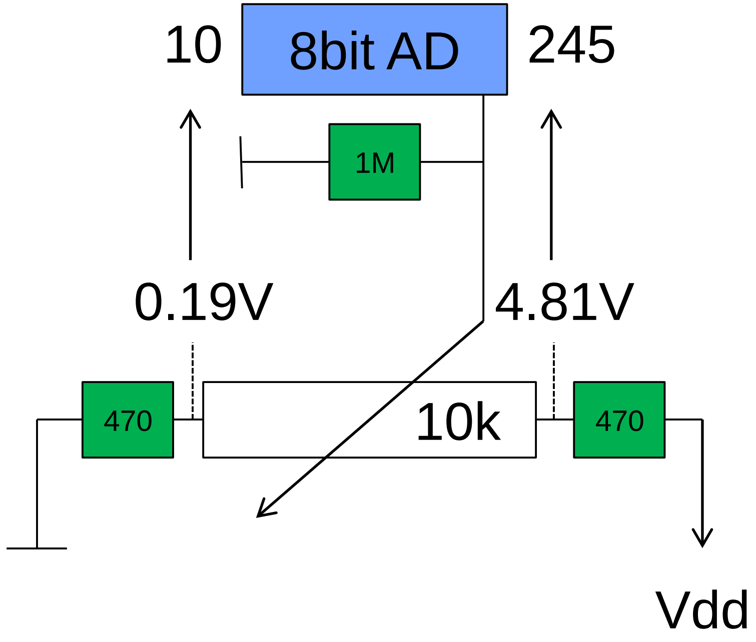
ASIL	Random hardware failure target values
D	$<10^{-8}\ h^{-1}$
C	$<10^{-7}\ h^{-1}$
B	$<10^{-7}\ h^{-1}$

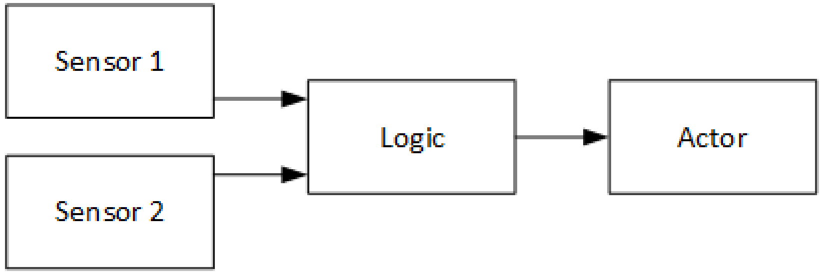
NOTE The quantitative target values described in this table can be tailored as specified in 4.2 to fit specific uses of the item (e.g. if the item is able to violate the safety goal for durations longer than the typical use of a passenger car).

Failure Rate of a Component versus Failure Rate of a Safety Function

- When talking about failure rates, we typically have to consider the complete safety function / channel
- I.e. the failure rates of components need to be added depending on the architecture

Range: 0..255
Valid Range: 11..244
Error Range: 0..10 and 245..255





$F_{S1+S2}(t) = F_{S1}(t) \times F_{S2}(t)$
$R_{S1+S2}(t) = R_{S1}(t) + R_{S2}(t) - R_{S1RS2}(t)$

Parallel components: Failure Rates are multiplied

$F_{Ch}(t) = F_{S1+S2}(t) + F_{L}(t) + F_{A}(t)$
$R_{ch}(t) = R_{S1+S2}(t) \times R_{L}(t) \times R_{A}(t)$

Serial components: Failure Rates are added

Failure Rate of a Component versus Failure Rate of a Safety Function
Example ISO 25119

Table B.1 — Example $MTTF_{DC}$ calculation of circuit board

Part number	Part description	$MTTF_i$ (from database) years	Dangerous failures %	$MTTF_{Di}$ years	$1/MTTF_{Di}$ 1/years	Number of parts	Total
1	Transistor, bipolar, low power	1 142	50	2 284	0,000 438	2	0,000 876
2	Resistor, carbon film	11 416	50	22 832	0,000 043 8	5	0,000 219
3	Capacitor, standard, no power	5 708	50	11 416	0,000 087 6	4	0,000 350
4	Relay (data coming from component manufacturer)	1 256	30	4 187	0,000 239	4	0,000 956
5	Contactor	32	20	160	0,006 25	1	0,006 25
$\sum(1/MTTF_{Di})$							0,008 65
$MTTF_{DC} = 1/\sum(1/MTTF_{Di})$ in years							115,6

This example gives an $MTTF_{DC}$ of 115,6 years.

B.4 Calculation of symmetric $MTTF_{DC}$ for two-channel architectures

Formula (B.4) is used to calculate a symmetric $MTTF_{DC}$ for a two-channel system (categories 3 and 4).

$$MTTF_{DC} = MTTF_{DC1} + MTTF_{DC2} - \frac{1}{\frac{1}{MTTF_{DC1}} + \frac{1}{MTTF_{DC2}}} \qquad (B.4)$$

where $MTTF_{DC1}$ and $MTTF_{DC2}$ are the values for two different redundant channels.

Formula (B.4) is not suitable for systems that are more complex. For further guidance in evaluating $MTTF_D$, see IEC 61508-6:2010, Annex B, IEC 62061 or use automated computational tools.

MTTF, MTTFd and MTTFdc

- Mean Time to Failure is the reciprocal values of the Failure Rate, expressed in years
- MTTFd is the percentage of dangerous failures
- And MTTFdc ist the resulting failure rate of a safety channel

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Hardware Failure Rate per Component

- The estimated failure rates for hardware parts used in the analyses shall be determined:
  - a) using hardware part failure rate data from a recognised industry source, or
  - b) using statistics based on field returns. In this case, the estimated failure rate should have a comparable confidence level of at least 70 %, or
  - c) using expert judgement founded on an engineering approach based on quantitative and qualitative arguments.

Acc. table 1: KFD2-PT2-Ex1-* Potentiometer converter 1oo1 structure

Parameters acc. to IEC61508	Variables
Device type	A
Demand mode	Low demand mode or High demand mode
Safety Function	Potentiometer converter
HFT	0
SIL	2
λ_{s}	0 FIT
λ_{dd}	0 FIT
λ_{du}	86.1 FIT
$\lambda_{fail\ high}$ [1]	6.35 FIT
$\lambda_{fail\ low}$ [1]	86.5 FIT
$\lambda_{no\ effect}$	218 FIT
$\lambda_{total\ (Safety\ function)}$	397 FIT
SFF	78.31%
MTBF [2]	265 years
PFH	$8.62 \cdot 10^{-8}$ 1/h
PFD_{avg} for T_1 = 1 year	$3.77 \cdot 10^{-4}$
$Tproof_{max}$	2.5 years

[1] "Fail high" and "fail low" failures are considered as dangerous detected failures.
[2] acc. To SN29500. This value includes failures which are not part of the safety function / MTTR = 8h

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PEPPERL+FUCHS	FMEDA – Report	respons.	DP.HSU	FS-0058PF-20A
	KFD2-PT2-Ex1-*	approved		
Mannheim		norm		sheet 4 of 10

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Hardware Failure Rate – SN29500 Failure Rates of Components

Part	Content
1	Expected values, General
2	Expected values for integrated circuits
3	Expected values for discrete semiconductors
4	Expected values for passive components
5	Expected values for electrical connections, electrical connectors and sockets
7	Expected values for relays
9	Expected values for switches and buttons
10	Expected values for signal and pilot lamps
11	Expected values for contactors
12	Expected values for optical components
15	Expected values for electromechanical protection devices in low voltage networks
16	Expected values for electromechanical Pushbuttons, Signaling Devices and Position Switches in Low Voltage Networks

Example Table SN29500

			Komplexität in Bit / *Complexity in bits*										$\theta_{vj,1}$ in °C
			512[1)] 16K	32K 64K	128K 256K	512K 1M	2M 4M	8M 16M	32M 64M	128M 256M	512M 1G	2G 4G	
			λ_{ref} in FIT										
Bipolar	**RAM, FIFO**	statisch *static*	50	60	-	-	-	-	-	-	-	-	75
	PROM		60	80	-	-	-	-	-	-	-	-	
MOS, CMOS, BICMOS	**RAM**	dynamisch *dynamic*	50	30	20	10	10	15	20	25		-	55
			-	-	-	-	-	-	-	-	70	(100)	70
	RAM, FIFO	statisch langsam >=30ns *static slow* 2)	15	10	10	10	10	30	50	-	-	-	55
		statisch schnell <30ns *static fast* 2)	30	25	15	25	40	55	90	-	-	-	70
	ROM *mask*		50	30	15	15	15	15	25	-	-	-	55
	EPROM, OTPROM UV-löschbar *UV eraseable*		30	30	20	20	20	20	40	-	-	-	
	FLASH		-	-	30	30	40	50	70	(100)	-	-	
			-	-	-	-	-	-	-	-	(200)	-	70
	EEPROM, EAROM		30	30	30	50	-	-	-	-	-	-	55

1 FIT=1x10^{-9} 1/h; (Ein Ausfall pro 10^{9} Bauelementestunden)

1 FIT equals one failure in 10^{9} component hours

Für Bauelemente ohne ausreichende Einsatzerfahrungen sind die Ausfallratenwerte eingeklammert.

Failure rates of components for which little operating experience has been gained are given in brackets.

Die Erfahrungswerte stammen von Speichern, in die nicht dauernd eingeschrieben bzw. von denen nicht dauernd gelesen wird.

The expected values have been gathered from memories which have not been written into or read from continuously.

Bei der Verwendung von nackten Speicherchips sind die Ausfallraten mit einem Faktor von bis zu 2 zu multiplizieren, wenn keine eigenen Erfahrungen in der Aufbautechnik vorliegen.

For bare memory chips the indicated failure rates shall be multiplied by a factor of up to two, if no first-hand experience in the design layout techniques is available.

1) Speicher mit < 512 Bit sind wie MSI-Familienschaltkreise entsprechender Technologie zu behandeln.

1) Memories < 512 bits are to be treated like MSI family circuits of the same technology.

2) Durch Bitfehler (soft errors) können bei statischen Speichern (SRAM) sporadische Ausfälle auftreten. Diese Ausfälle sind durch Fehlerkorrektur oder Neuladen behebbar. Für sporadische Ausfälle muss mit einer Ausfallrate bis 1000 FIT/Mbit gerechnet werden (abhängig von der Applikation und der Halbleiter-Technologie).

2) Sporadic failures may occur in static memories (SRAM) caused by bit errors (soft errors). These failures can be removed by Error Correction Circuits or reloading. For sporadic failures, a failure rate of up to 1000 FIT/Mbit can be expected (dependent on application and semiconductor technology).

Die Ausfallrate bei Betriebsbedingungen λ berechnet sich zu

The failure rate under operating conditions λ is calculated as follows

$$\lambda = \lambda_{ref} \times \pi_U \times \pi_I \times \pi_T \quad (1.1)$$

Hierin bedeuten / *where*:

λ_{ref}	Ausfallrate bei Referenzbedingungen	failure rate under reference conditions
π_U	Faktor für Spannungsabhängigkeit	voltage dependence factor
π_I	Faktor für Stromabhängigkeit	current dependence factor
π_T	Faktor für Temperaturabhängigkeit	temperature dependence factor

Tabelle 2 Ausfallraten für Widerstände
Table 2 Failure rates for resistors

Widerstand / *Resistor*		λ_{ref} in FIT	θ_1 [1)] in °C
Kohleschicht / *Carbon film*	≤100 kOhm >100 kOhm	0,3 1	55
Metallschicht / *Metal film*		0,2	55
Netzwerke (Schichtschaltung) je Widerstandselement ***Networks (film circuits) per resistor element***	Standard kundenspezifische / *Custom design*	0,1 0,5	55
Metalloxidschicht / *Metal-oxide*		5	85
Draht / *Wire-wound*		5	85
Veränderbare / *Variable*		30	55

1 FIT = 1x10^{-9} 1/h (ein Ausfall pro 10^{9} Bauelementestunden)
1) Oberflächentemperatur

1 FIT equals one failure per 10^{9} component hours
1) Resistor element temperature

Diagnostic Coverage

This requirement applies to ASIL (B), C, and D of the safety goal. If the evidence that can be made available for the **hardware part failure rate is insufficient**, then alternative means shall be proposed (e.g. add safety mechanisms to detect and control this fault). If the alternative means consist solely of added safety mechanisms:

- a) the diagnostic coverage of the hardware part with respect to residual faults shall be equal to or higher than the item SPFM target value; and
- b) the diagnostic coverage of the hardware part with respect to latent faults shall be equal to or higher than the item LFM target value.

Table 4 — Possible source for the derivation of the target "single-point fault metric" value

	ASIL B	ASIL C	ASIL D
Single-point fault metric	≥90 %	≥97 %	≥99 %

Table 5 — Possible source for the derivation of the target "latent-fault metric" value

	ASIL B	ASIL C	ASIL D
Latent-fault metric	≥60 %	≥80 %	≥90 %

Design Pattern for DC (selection)

Table D.2 — E/E Systems

Safety mechanism/ measure	See overview of techniques	Typical diagnostic coverage considered achievable	Notes
Failure detection by on-line monitoring	D.2.1.1	Low	Depends on diagnostic coverage of failure detection
Comparator	D.2.1.2	High	Depends on the quality of the comparison
Majority voter	D.2.1.3	High	Depends on the quality of the voting
Dynamic principles	D.2.2.1	Medium	Depends on diagnostic coverage of failure detection
Analogue monitoring of digital signals	D.2.2.2	Low	—
Self-test by software cross exchange between two independent units	D.2.3.3	Medium	Depends on the quality of the self-test

Table D.6 — Communication bus (serial, parallel)

Safety mechanism/ measure	See overview of techniques	Typical diagnostic coverage considered achievable	Notes
One-bit hardware redundancy	D.2.5.1	Low	—
Multi-bit hardware redundancy	D.2.5.2	Medium	—
Read back of sent message	D.2.5.9	Medium	—
Complete hardware redundancy	D.2.5.3	High	Common mode failures can reduce diagnostic coverage
Inspection using test patterns	D.2.5.4	High	—
Transmission redundancy	D.2.5.5	Medium	Depends on type of redundancy. Effective only against transient faults
Information redundancy	D.2.5.6	Medium	Depends on type of redundancy
Frame counter	D.2.5.7	Medium	—
Timeout monitoring	D.2.5.8	Medium	—
Combination of information redundancy, frame counter and timeout monitoring	D.2.5.6, D.2.5.7 and D.2.5.8	High	For systems without hardware redundancy or test patterns, high coverage can be claimed for the combination of these safety mechanisms

Table D.5 — Analogue and digital I/O

Safety mechanism/ measure	See overview of techniques	Typical diagnostic coverage considered achievable	Notes
Failure detection by on-line monitoring (Digital I/O)[a]	D.2.1.1	Low	Depends on diagnostic coverage of failure detection
Test pattern	D.2.4.1	High	Depends on type of pattern
Code protection for digital I/O	D.2.4.2	Medium	Depends on type of coding
Multi-channel parallel output	D.2.4.3	High	—
Monitored outputs	D.2.4.4	High	Only if dataflow changes within diagnostic test interval
Input comparison/ voting (1oo2, 2oo3 or better redundancy)	D.2.4.5	High	Only if dataflow changes within diagnostic test interval

[a] Digital I/O can be periodic.

Table D.4 — Processing units

Safety mechanism/ measure	See overview of techniques	Typical diagnostic coverage considered achievable	Notes
Self-test by software: limited number of patterns (one channel)	D.2.3.1	Medium	Depends on the quality of the self-test
Self-test by software cross exchange between two independent units	D.2.3.3	Medium	Depends on the quality of the self-test
Self-test supported by hardware (one-channel)	D.2.3.2	Medium	Depends on the quality of the self-test
Software diversified redundancy (one hardware channel)	D.2.3.4	High	Depends on the quality of the diversification. Common mode failures can reduce diagnostic coverage
Reciprocal comparison by software	D.2.3.5	High	Depends on the quality of the comparison
HW redundancy (e.g. dual core lockstep, asymmetric redundancy, coded processing)	D.2.3.6	High	It depends on the quality of redundancy. Common mode failures can reduce diagnostic coverage
Configuration register test	D.2.3.7	High	Configuration registers only
Stack over/under flow Detection	D.2.3.8	Low	Stack boundary test only
Integrated hardware consistency monitoring	D.2.3.9	High	Coverage for illegal hardware exceptions only

NOTE This table deals only with safety mechanisms dedicated to processing units. General techniques like one based on data comparison (see D.2.1.2) are also able to detect failures of electrical elements but are not integrated in this table (already included in Table D.2 — E/E Systems).

Design Verification - ISO 26262 Fault Classification

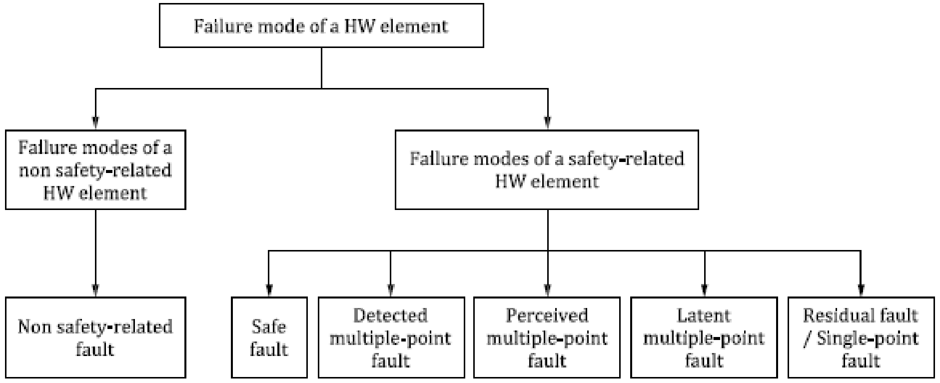


Figure B.1 — Failure mode classifications of a hardware element

- Single Point Fault
  hardware fault in an element that leads directly to the violation of a safety goal
- Residual Fault
  portion of a random hardware fault that by itself leads to the violation of a safety goal, where that portion of the random hardware fault is not controlled by a safety mechanism
- Multiple Point Fault
  individual fault that, in combination with other independent faults, if undetected and not perceived, could lead to a multiple-point failure
- Perceived Fault
  fault that may be perceived indirectly (through deviating behaviour on vehicle level)
- Safe Fault
  fault whose occurrence will not significantly increase the probability of violation of a safety goal

Failure Mode Classification Workflow

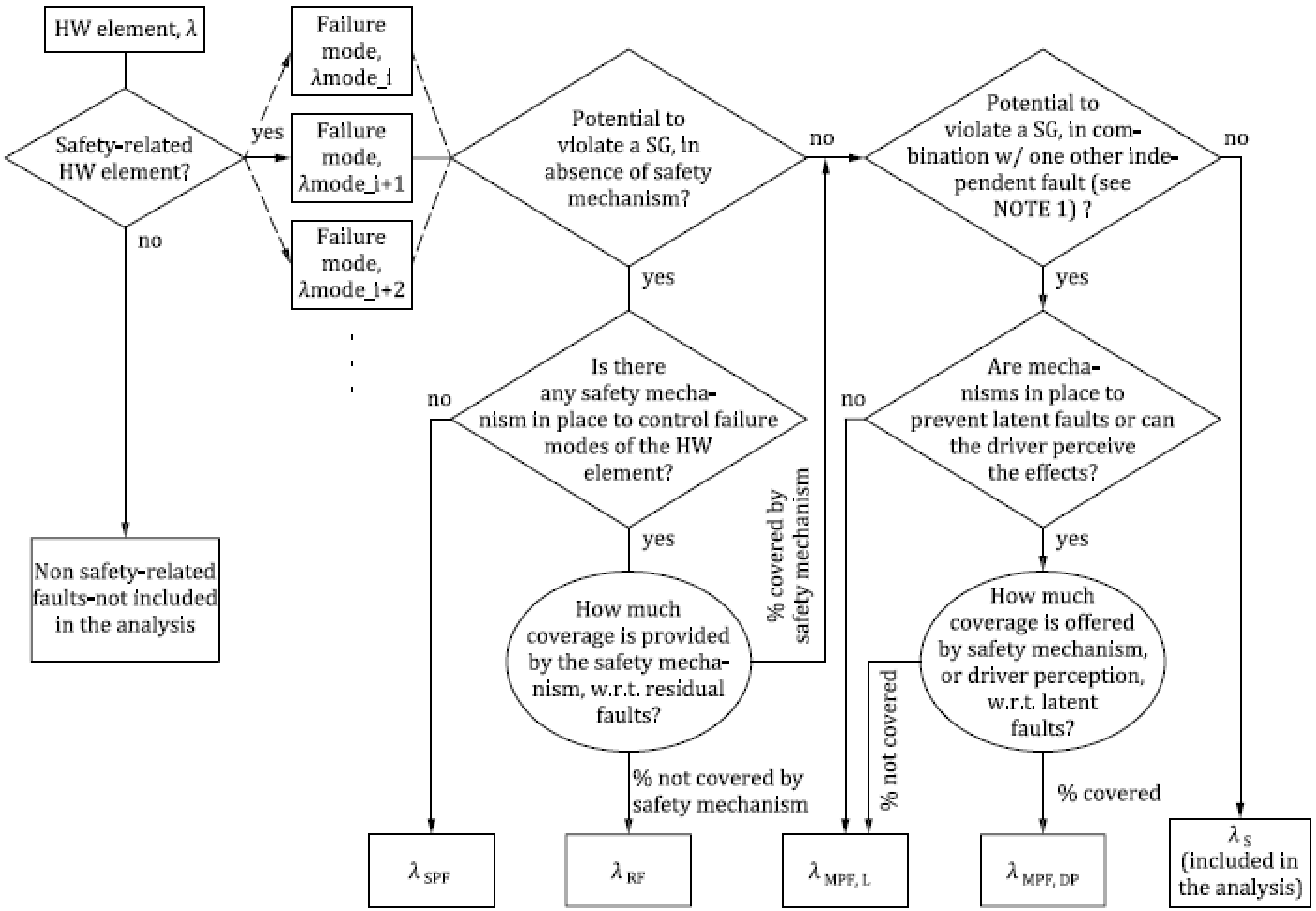


Figure B.2 — Example of flow diagram for failure mode classification

Fault Metrics – Single Point fault Metric

Considering the overall failure rate of an item

$$\lambda = \lambda_{SPF} + \lambda_{RF} + \lambda_{MPF} + \lambda_{S}$$

The Single-point fault metric reflects the robustness of the item to single-point and residual faults either by coverage from safety mechanisms or by design (primarily safe faults). A high single-point fault metric implies that the proportion of single-point faults and residual faults in the hardware of the item is low.

$$SPFM = 1 - \frac{\sum_{SR\ HW}(\lambda_{SPF} + \lambda_{RF})}{\sum_{SR\ HW}\lambda}$$

Fault Metrics – Latent Point fault Metric

Considering the overall failure rate of an item

$$\lambda = \lambda_{SPF} + \lambda_{RF} + \lambda_{MPF} + \lambda_{S}$$

This metric reflects the robustness of the item to latent faults either by coverage of faults in safety mechanisms or by the driver recognizing that the fault exists before the violation of the safety goal, or by design (primarily safe faults). A high latent-fault metric implies that the proportion of latent faults in the hardware is low.

$$LPFM = 1 - \frac{\sum_{SR\ HW}\left(\lambda_{MPF,L}\right)}{\sum_{SR\ HW}(\lambda - \lambda_{SPF} - \lambda_{RF})}$$

Example Calculation SPFM

Function 1 has one input (temperature measured via sensor R3) and one output (valve 2 controlled by I71) and its behaviour is to open valve 2 when the temperature is higher than 90 °C.

If no current flows through I71, valve 2 is open.

The associated safety goal 1 is “valve 2 shall not be closed for longer than 100 ms when the temperature is higher than 100 °C”. The safety goal is assigned ASIL B. The safe state is: valve 2 open.

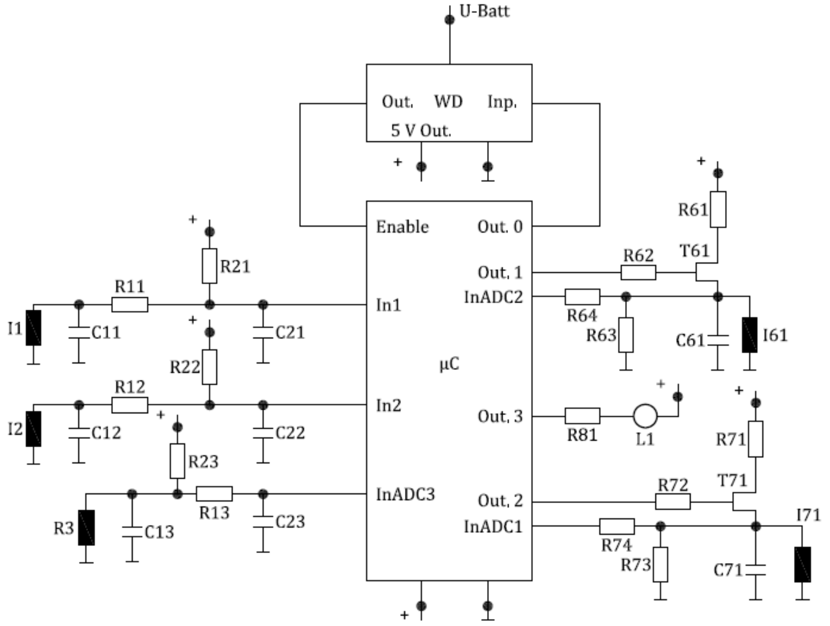


Figure E.1 — Example diagram

Table 4 — Possible source for the derivation of the target “single-point fault metric” value

	ASIL B	ASIL C	ASIL D
Single-point fault metric	≥90 %	≥97 %	≥99 %

Component Name	Failure rate/ FIT	Safety-related component to be considered in the calculations?	Failure Mode	Failure mode distribution	Failure mode that has the potential to violate the safety goal in absence of safety mechanisms?	Safety mechanism(s) allowing to prevent the failure mode from violating the safety goal?	Failure mode coverage wrt. violation of safety goal	Residual or Single-Point Fault failure rate/ FIT	Failure mode that may lead to the violation of safety goal in combination with an independent failure of another component?	Detection means? Safety mechanism(s) allowing to prevent the failure mode from being latent?	Failure mode coverage with respect to latent failures	Latent Multiple-Point Fault failure rate/ FIT
R3 NOTE 1	3	YES	open	30 %	X	none	0 %	0,9				
			closed	10 %								
			drift 0,5	30 %								
			drift 2	30 %	X		0 %	0,9				
R13 NOTE 1, NOTE 2 and NOTE 6	2	YES	open	90 %	X	none	0 %	1,8				
			closed	10 %	X		0 %	0,2				
R23 NOTE 1	2	YES	open	90 %		none						
			closed	10 %	X		0 %	0,2				
C13 NOTE 3 and NOTE 6	2	YES	open	20 %	X	none	0 %	0,4				
			closed	80 %								
C23 NOTE 4	2	NO	open	20 %								
			closed	80 %								
WD	20	YES	Out. Stuck at 1	50 %					X	none	0 %	10
			Out. Stuck at 0	50 %								
T71 NOTE 5	5	YES	open circuit	50 %		SM1				SM1		
			short circuit	50 %	X		90 %	0,25	X		80 %	0,45
R71 NOTE 2 and NOTE 6	2	YES	open	90 %						none		
			closed	10 %					X		0 %	0,2
R72 NOTE 2 and NOTE 6	2	YES	open	90 %						none		
			closed	10 %					X		0 %	0,2
R73 NOTE 4	2	NO	open	90 %								
			closed	10 %								
R74 NOTE 2 and NOTE 6	2	YES	open	90 %					X	none	0 %	1,8
			closed	10 %					X		0 %	0,2
I71 NOTE 4	5	NO	open	80 %								
			closed	20 %								
C71 NOTE 3	2	YES	open	20 %					X	none		0,4
			closed	80 %								
R81 NOTE 4	2	NO	open	90 %								
			closed	10 %								
L1 NOTE 4	10	NO	open	90 %								
			closed	10 %								
μC	100	YES	All	50 %	X	SM4	90 %	5	X	SM4	100 %	0
			All	50 %								
							∑ 9,65				∑ 13,25	

Total failure rate 163 FIT

Total Safety-Related 142 FIT

Total Non Safety-Related 21 FIT

Single-Point Fault Metric = 1-(9,65/142) = 93,2 %

Latent Fault Metric = 1-(13,25/(142-9,65)) = 90,0 %

Table 5 — Possible source for the derivation of the target “latent-fault metric” value

	ASIL B	ASIL C	ASIL D
Latent-fault metric	≥60 %	≥80 %	≥90 %

Checklist Hardware Failure Modes

Electrical Elements
- Open circuit
- Contact resistance
- Short circuits to Vbat and GND
- Short circuit between neighboring pins
- Resistive drift between pins

Sensors incl. switches
- Out of range
- Offsets
- Stuck in Range
- Oscillations

Actuators
- No generic points mentioned, individual analysis

Power supply
- Drift and oscillations
- Under and Over Voltage
- Power spikes

Clocks
- Incorrect frequency
- Jitter

Checklist Microcontroller (extract)

Selftest , HW Diagnostics

Table 30 — Example of failure modes for digital components

Part/subpart	Function	Aspects to be considered for Failure mode[a]
Central Processing Unit (CPU)	Execute given instruction flow according to given Instruction Set Architecture.	CPU_FM1: given instruction flow(s) not executed (total omission) CPU_FM2: un-intended instruction(s) flow executed (commission) CPU_FM3: incorrect instruction flow timing (too early/late) CPU_FM4: incorrect instruction flow result CPU_FM1 can be further refined if necessary into: — CPU_FM1.1: given instruction flow(s) not executed (total omission) due to program counter hang up — CPU_FM1.2: given instruction flow(s) not executed (total omission) due to instruction fetch hang up
CPU Interrupt Handler circuit (CPU_INTH)	Execute interrupt service routine (ISR) according to interrupt request	CPU_INTH_FM1: ISR not executed (omission/too few) CPU_INTH_FM2: un-intended ISR execution (commission/too many) CPU_INTH_FM3: delayed ISR execution (too early/late) CPU_INTH_FM4: incorrect ISR execution (see CPU_FM1/2/4)
CPU Memory Management Unit (CPU_MMU)	The Memory Management Unit (MMU) typically performs two functions: — translates virtual addresses into physical addresses — Controls memory access permissions.	CPU_MMU_FM1: Address translation not executed CPU_MMU_FM2: Address translation when not requested CPU_MMU_FM3: delayed address translation CPU_MMU_FM4: translation with incorrect physical address CPU_MMU_FM5: un-intended blocked access CPU_MMU_FM6: un-intended allowed access CPU_MMU_FM7: delayed access
Interrupt Controller Unit (ICU)	Send interrupt requests to given CPU according to hardware-based or software-based interrupt events and according to intended quality of service (e.g. priority). The interrupt controller can service multiple CPUs.	ICU_FM1: Interrupt request to CPU missing ICU_FM2: Interrupt request to CPU without triggering event ICU_FM3: Interrupt request too early/late ICU_FM4: Interrupt request sent with incorrect data
DMA	Data Transfer: Move data when requested from source address(es) to destination address(es) and notify the data transfer completion. The set of data transferred is called a message.	DMA_FM1: No requested data transfer. The message is not sent as intended to the destination address. DMA_FM2: Data transfer without a request. DMA_FM3: Data transfer too early/late. DMA_FM4: Incorrect output

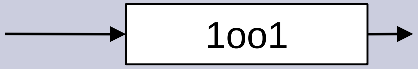
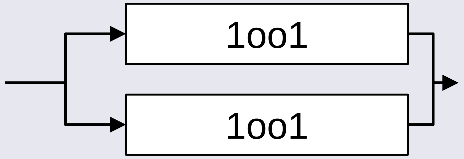
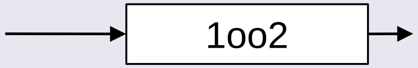
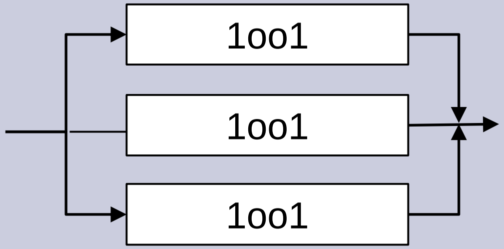
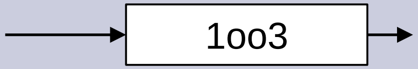
[a] Failure Modes can be caused by permanent and transient random hardware faults.

Table 30 *(continued)*

Part/subpart	Function	Aspects to be considered for Failure mode[a]
(first level of abstraction)		
Buses and Interconnects (internal communication)	Deliver bus transaction initiated from a given bus master to the target address according to the intended quality of service (TXFR). A transaction is a given set of data as defined by the bus protocol.	BUS_TXFR_FM1: Requested transaction not delivered BUS_TXFR_FM2: Transaction delivered without a request BUS_TXFR_FM3: Transaction delivered with incorrect timing BUS_TXFR_FM3: Transaction delivered with incorrect data
External SDRAM with SDRAM Controller	Volatile memory fetch (read) or store (write) data to given row and column address according to input command from SDRAM controller.	SDRAM_RW_FM1: given write/read access not executed (omission) SDRAM_RW_FM2: un-intended write/read access executed (commission) SDRAM_RW_FM3: incorrect write/read access result (too early/late) SDRAM_RW_FM4: incorrect write/read access result
or (second level of abstraction)		
External SDRAM with SDRAM Controller	SDRAM controller provides row address to be prepared for read or write operation on a selected bank.	SDRAM_RA_FM1: given row address not accessed (omission) SDRAM_RA_FM2: un-intended row address accessed (commission) SDRAM_RA_FM3: delayed row address result (too early/late) SDRAM_RA_FM4: incorrect row address result
External SDRAM with SDRAM Controller	SDRAM controller provides column address to access data for read or write operation.	SDRAM_CA_FM1: given column address not accessed (omission) SDRAM_CA_FM2: un-intended column address accessed (commission) SDRAM_CA_FM3: delayed column address result (too early/late) SDRAM_CA_FM4: incorrect column address result
External SDRAM with SDRAM Controller	SDRAM controller provides commands (e. g. activate, write, read, pre-charge, refresh ...) to get data for read or write operation.	SDRAM_IN_FM1: given instruction not executed (omission) SDRAM_IN_FM2: un-intended instruction executed (commission) SDRAM_IN_FM3: delayed instruction result (too early/late) SDRAM_IN_FM4: incorrect instruction result
External SDRAM with SDRAM Controller	SDRAM data path provides write/read data to/from memory array.	SDRAM_DW_FM1: given data word not executed (omission) SDRAM_DW_FM2: un-intended data word executed (commission) SDRAM_DW_FM3: delayed data word result (too early/late) SDRAM_DW_FM4: incorrect data word result

[a] Failure Modes can be caused by permanent and transient random hardware faults.

Slightly different strategy in IEC61508

Structure		Hardware Fault Tolerance
		0 A single fault can cause a failure
		1 Two or more faults can cause a failure
		2 Three or more faults can cause a failure

IEC61508 – Additional aspect of structural reliability

Based on the failure rate of individual components, the highest possible SIL is depending on the architecture

SFF	HFT 0	HFT 1	HFT 2
< 60%	SIL 1	SIL 2	SIL 3
60% - 90%	SIL 2	SIL 3	SIL 4
90% - 99%	SIL 3	SIL 4	SIL 4
>= 99%	SIL 3	SIL 4	SIL 4

If the failure rate of one or more components cannot be determined, the following table is valid

SFF	HFT 0	HFT 1	HFT 2
< 60%	-/-	SIL 1	SIL 2
60% - 90%	SIL 1	SIL 2	SIL 3
90% - 99%	SIL 2	SIL 3	SIL 4
>= 99%	SIL 3	SIL 4	SIL 4

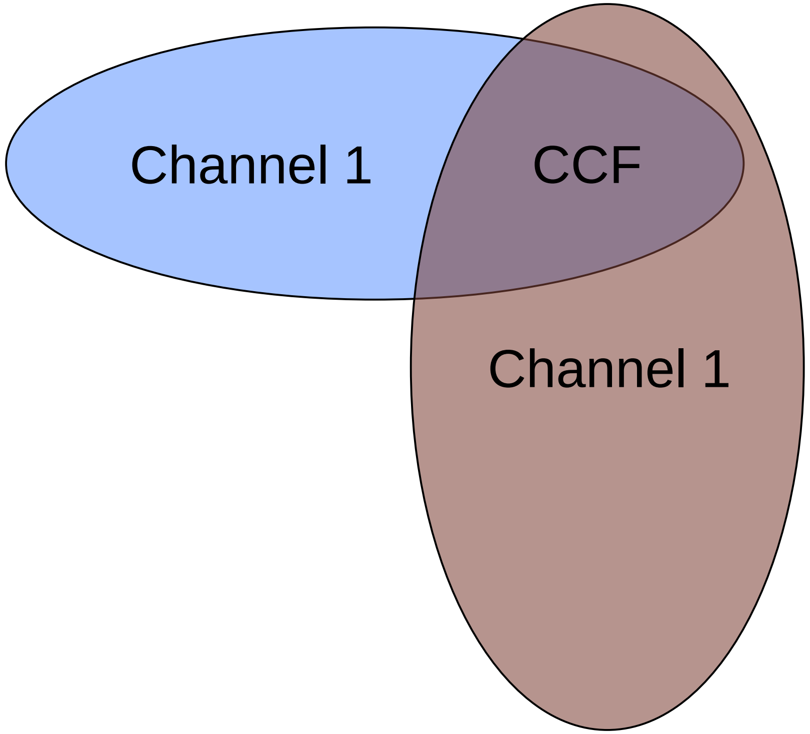
Embedded

Common Cause Failure

- In case of redundant systems, a common cause failure analysis is required, which systematically searches for errors which have an impact on more than one channel.

- Typical causes
  - Power supply
  - EMD
  - Temperature
  - Technology

- β – CCF factor
  - Amount of not detected failures caused by CCF



CCF calculation
Example IEC 61508 Part 6.D

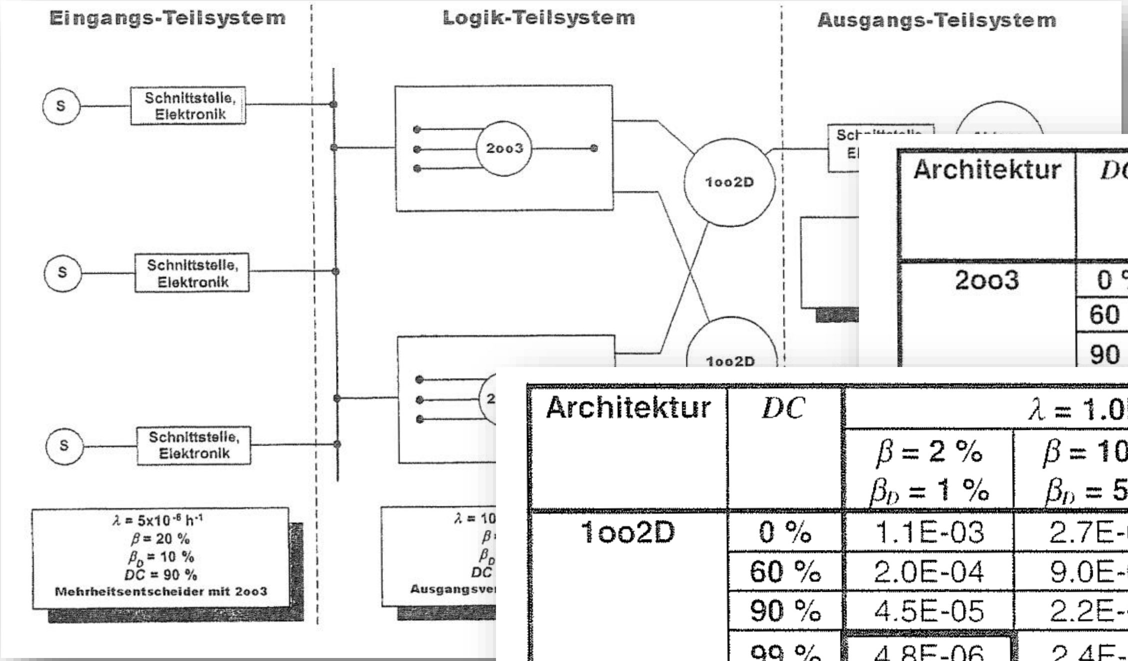
Tabelle D.1 – Anrechnung programmierbarer Elektronik oder Sensoren/Stellglieder

Merkmal	Logik-Teilsystem		Sensoren und Stellglieder	
	X_{LS}	Y_{LS}	X_{SI}	Y_{SI}
Trennung				
Werden alle Signalkabel an allen Stellen für die Kanäle getrennt geführt?	1.5	1.5	1.0	2.0
Sind die Kanäle des Logik-Teilsystem auf getrennten Leiterpiatten?	3.0	1.0		
Sind die Kanäle des Logik-Teilsystem in getrennten Schaltschränken?	2.5	0,5		
Wenn die Sensoren/Stellglieder zugeordnete Steuerelektroniken haben. befinden sich diese für jeden Kanal auf getrennten Leiterplatten?			2.5	1,5
Wenn die Sensoren/Stellglieder zugeordnete Steuerelektroniken haben. befinden sich diese füi· jeden Kanal im Innenraum und in getrennten Schaltschränken?			2,5	0.5
Diversität/Redundanz				
Verwenden die Kanäle verschiedene Technoiogien. z. B. einer Elekrronik oder programmierbare Elektronik und der andere Relais?	7,0			
Verwenden die Kanäle verschiedene elektronische Technologien. z. B. einer Elektronik. der andere programmierbare Elektronik?	5,0			
Verwenden die Sensoren verschiedene physikalische Prinzipien, z. B. Druck und Temperatur, Messung der Strömungsgeschwindigkeit mit einem Fiügelrad-Aufinehmer oder mit dem Doooler-Effekt?			7,5	
Verwenden die Geräte andere elektrische Prinzipien oder Konstruktionen. z. B. digital und analog, verschiedene Hersteller (nicht nur baugleiches Gerät mit			5,5	

Calculation of the System Reliability

Based on
- Component failure rate
- Architecture /HTF, SFF, DC and CCF)

The overall reliability can be calculated. Stochastic approaches are e.g. described in IEC61508 Part 6.B



Architektur	DC	λ = 5.0E-06		
		β = 2 % β_D = 1 %	β = 10 % β_D = 5 %	β = 20 % β_D = 10 %
2oo3	0 %	6.8E-04	1.5E-03	2.5E-03
	60 %	1.6E-04	5.1E-04	9.4E-04
	90 %	2.7E-05	1.2E-04	2.3E-04
				2.4E-05

... Auszug aus

Architektur	DC	λ = 1.0E-05		
		β = 2 % β_D = 1 %	β = 10 % β_D = 5 %	β = 20 % β_D = 10 %
1oo2D	0 %	1.1E-03	2.7E-03	4.8E-03
	60 %	2.0E-04	9.0E-04	1.8E-03
	90 %	4.5E-05	2.2E-04	4.4E-04
	99 %	4.8E-06	2.4E-05	4.8E-05

ANMERKUNG Diese Tabelle ist ein Auszug aus Tabelle B.3.

Für das Eingangs-Teilsystem:

$$PFD_S = 2{,}3 \times 10^{-4}$$

Für das Logik-Teilsystem:

$$PFD_L = 4{,}8 \times 10^{-6}$$

Für das Ausgangs-Teilsystem:

$$PFD_{FE} = 4{,}4 \times 10^{-3} + 8{,}8 \times 10^{-3} = 1{,}3 \times 10^{-2}$$

Daraus ergibt sich für die Sicherheitsfunktion:

$$PFD_{SYS} = 2{,}3 \times 10^{-4} + 4{,}8 \times 10^{-6} + 1{,}3 \times 10^{-2} = 1{,}3 \times 10^{-2}$$

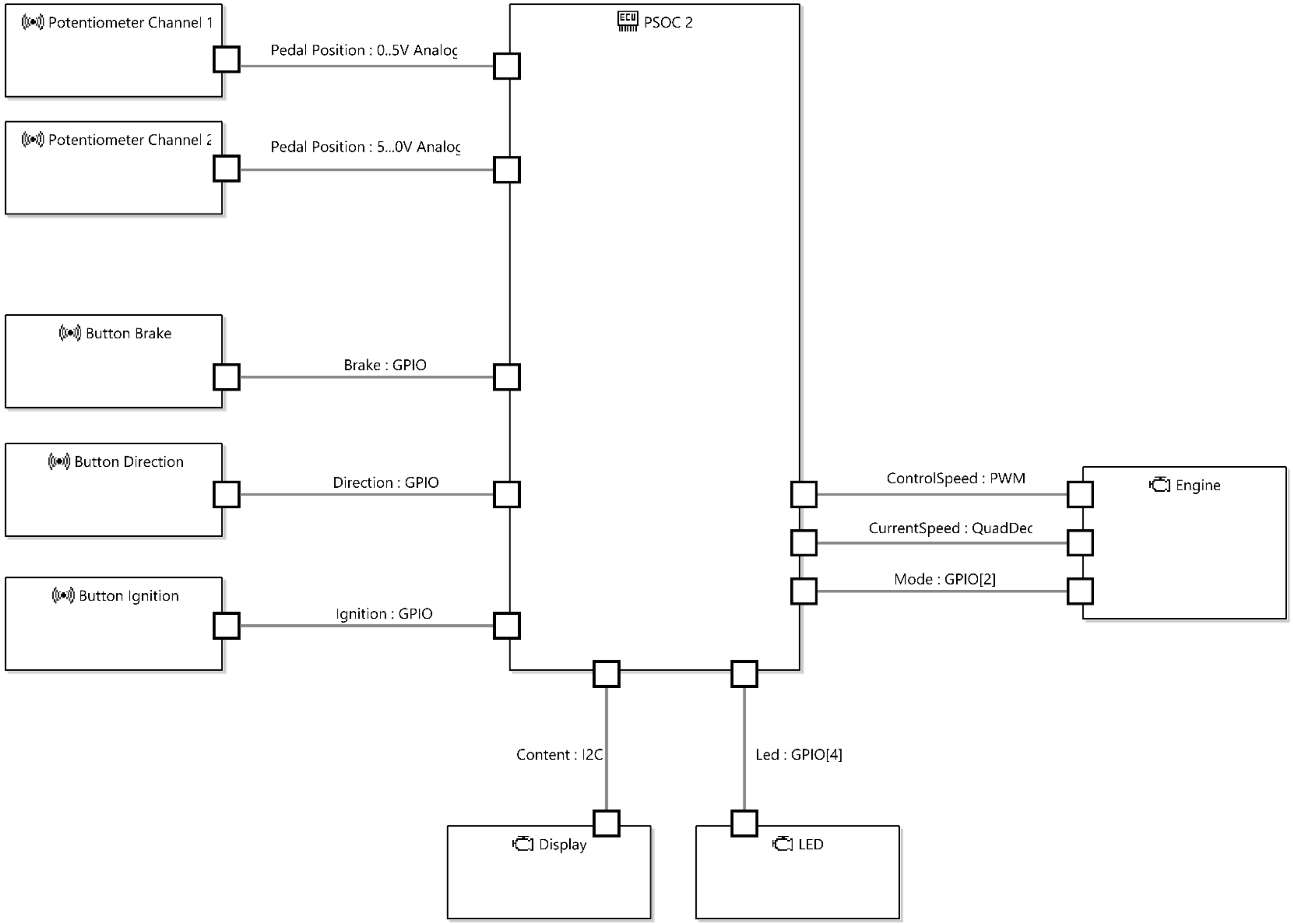
$\equiv$ **SIL (Sicherheits-Integritätslevel) 1**

Hardware Design

Let’s create the hardware architecture.

And assess it!

First version: Single PSOC with redundant sensors



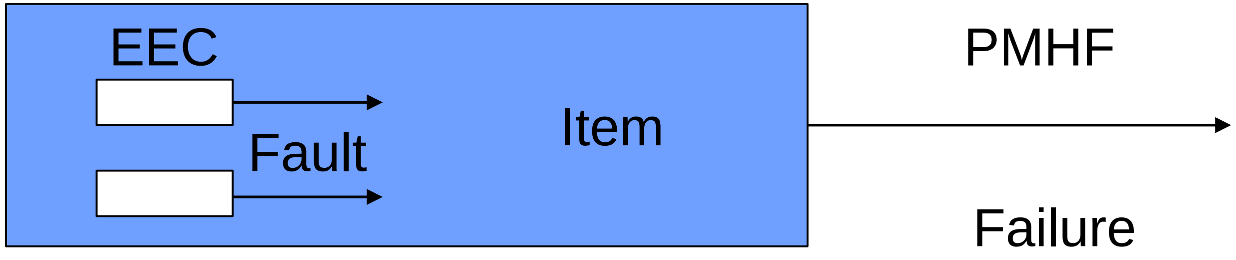
Alternative Analysis / Assessment methods

Option 1: Evaluation of Probabilistic Metric for random Hardware Failures (PMHF)

- represents a quantitative analysis which evaluates the violation of the considered safety goal by random failures of the hardware elements. The quantitative result of the analysis is compared with a target value.
- Focus on **failure** of the complete item (example: engine ECU, where historical data is available by reusing it from a previous project or vendor specific data)

Option 2: Evaluation of Each Cause of safety goal violation (EEC)

- is based on the individual evaluation of each hardware part and its contribution to the violation of the considered safety goal with respect to residual, single-point and plausible dual-point failures
- Focus on individual **faults** and resulting failures of the item



PMHF Workflow

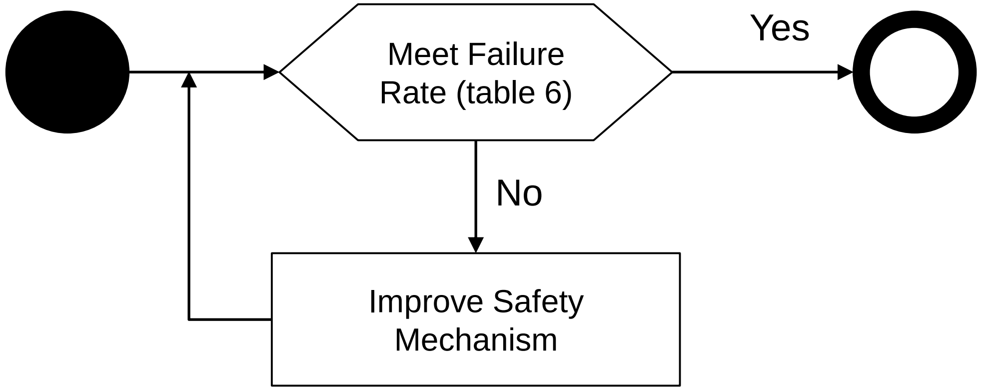


Table 6 — Possible source for the derivation of the random hardware failure target values

ASIL	Random hardware failure target values
D	$<10^{-8}\ h^{-1}$
C	$<10^{-7}\ h^{-1}$
B	$<10^{-7}\ h^{-1}$
NOTE The quantitative target values described in this table can be tailored as specified in 4.2 to fit specific uses of the item (e.g. if the item is able to violate the safety goal for durations longer than the typical use of a passenger car).	

EEC - Analysis workflow for single point and residual failures

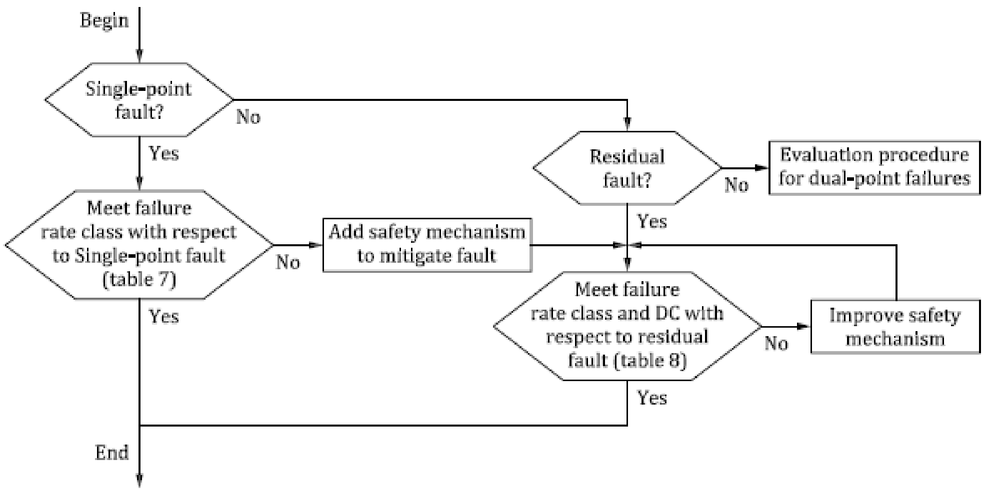


Figure 3 — Evaluation procedure for single-point and residual faults

Failure rate class	Failure rate classes target values	
1	< PFH (ASIL D) / 100	$< 10^{-10}$ / h
2	< PFH (ASIL D) / 10	$< 10^{-9}$ / h
3	< PFH (ASIL D)	$< 10^{-8}$ / h
4	< PFH (ASIL D) x 10	$< 10^{-7}$ / h
5	< PFH (ASIL D) x 100	$< 10^{-6}$ / h

Table 7 — Targets of failure rate classes of hardware parts regarding single-point faults

ASIL of the safety goal	Failure rate class
D	Failure rate class 1 + dedicated measures[a]
C	Failure rate class 2 + dedicated measures[a] or Failure rate class 1
B	Failure rate class 2 or Failure rate class 1

[a] The NOTE 2 in requirement 9.4.1.2 gives examples of dedicated measures.

Table 8 — Maximum failure rate classes for a given diagnostic coverage of the hardware part — residual faults

ASIL of the safety goal	Diagnostic coverage with respect to residual faults			
	≥99,9 %	≥99 %	≥90 %	<90 %
D	Failure rate class 4	Failure rate class 3	Failure rate class 2	Failure rate class 1 + dedicated measures[a]
C	Failure rate class 5	Failure rate class 4	Failure rate class 3	Failure rate class 2 + dedicated measures[a]
B	Failure rate class 5	Failure rate class 4	Failure rate class 3	Failure rate class 2

[a] The NOTE 2 in requirement 9.4.1.2 gives examples of dedicated measures.

NOTE 2 Table 8 specifies the connection between maximum failure rate class allowed given the target ASIL and the diagnostic coverage. Lower failure rate classes are acceptable, but not required.

NOTE 3 "Lower failure rate classes" means failure rate classes with a lower number. For example, "lower failure rate classes" with respect to failure rate class 3 means failure rate classes 2 and 1.

NOTE 4 The proportion of safe faults of the hardware part can be considered when determining the coverage of the safety mechanisms. In this case, the calculation of the coverage is done analogously to the calculation of the single-point fault metric, but at the hardware part level instead of at the item level.

EEC for dual-point failures

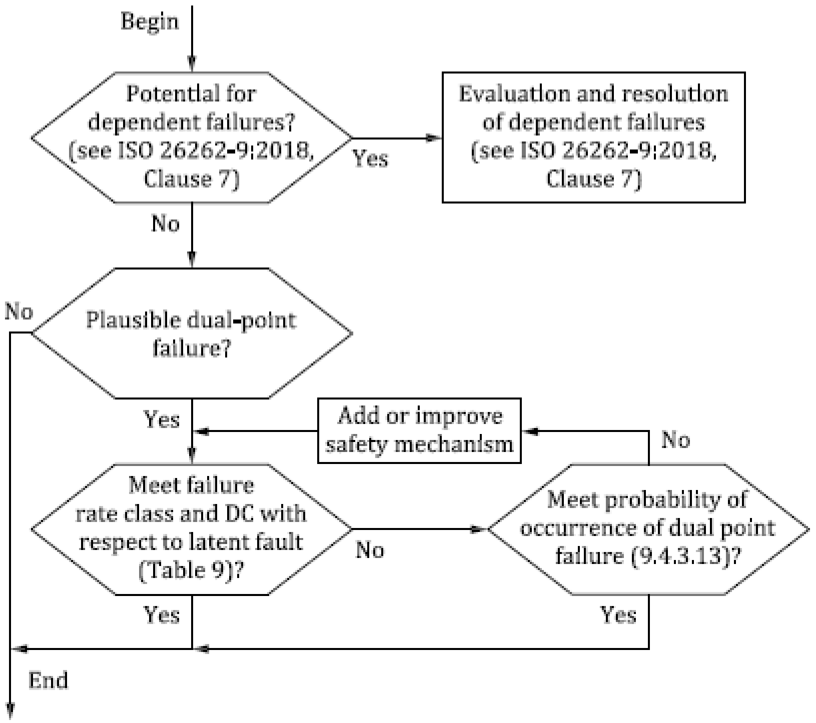


Figure 4 — Evaluation procedure for dual-point failures

- Only relevant for ASIL C and D
- Focus on dual-point failures
- Higher (triple etc) multi-point failures only considered if relevant for the safety goal

Table 9 — Targets of failure rate class and coverage of hardware part regarding dual-point faults

ASIL of safety goal	Diagnostic coverage with respect to latent faults		
	≥99 %	≥90 %	<90 %
D	Failure rate class 4	Failure rate class 3	Failure rate class 2
C	Failure rate class 5	Failure rate class 4	Failure rate class 3

One Pager ISO26262

	Residual Risk for Safety Goal Failure Level - - - - - - - - - - - -	*Alternative:* Residual Risk for Safety Goal Fault Level - - - - - -								Effectiveness of the system architecture	
ASIL	Random hardware failure target values	Single Point Faults	Residual Faults				Latent Multiple Point Faults			SPFM	LFM
	Probability of failure per hour (PFH)		Diagnostic Coverage (DC)				Diagnostic Coverage (DC)			Single Point Fault Metric	Latent Fault Metric
			≥ 99,9%	≥ 99%	≥ 90%	< 90%	≥ 99%	≥ 90%	< 90%		
A	--	--		--	--	--	--	--	--	--	--
B	Recommended: $< 10^{-7}$ / h (100 FIT)	2 OR 1	5	4	3	2	--	--	--	≥ 90%	≥ 60%
C	Requirement: $< 10^{-7}$ / h (100 FIT)	2+ OR 1	5	4	3	2+	5	4	3	≥ 97%	≥ 80%
D	Requirement: $< 10^{-8}$ / h (10 FIT)	1+	4	3	2	1+	4	3	2	≥ 99%	≥ 90%

Note: 1+, 2+ means failure rate class 1 or 2 in addition with dedicated measures such as burn-in test, special characteristic etc. acc. to ISO 26262 part 5 ch. 9.4.2.4

Failure rate class	Failure rate classes target values	
1	< PFH (ASIL D) / 100	$< 10^{-10}$ / h
2	< PFH (ASIL D) / 10	$< 10^{-9}$ / h
3	< PFH (ASIL D)	$< 10^{-8}$ / h
4	< PFH (ASIL D) x 10	$< 10^{-7}$ / h
5	< PFH (ASIL D) x 100	$< 10^{-6}$ / h

$$SPFM = 1 - \frac{\lambda_{SPF} + \lambda_{RF}}{\lambda_{total}} = \frac{\lambda_{MPF} + \lambda_{S}}{\lambda_{total}}$$

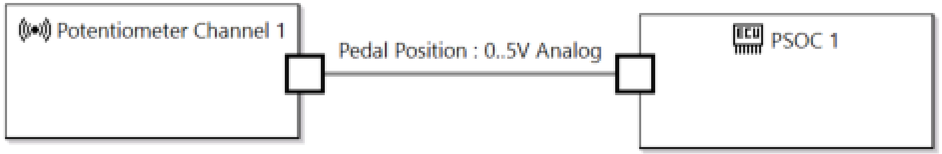
$$\lambda_{RF} = \lambda_{total} \times \left(1 - \frac{DC_{RF}}{100}\right)$$

$$LFM = 1 - \frac{\lambda_{MPF\ Latent}}{\lambda_{MPF} + \lambda_{S}} = \frac{\lambda_{MPF\ Perceived} + \lambda_{MPF\ Detected} + \lambda_{S}}{\lambda_{MPF} + \lambda_{S}}$$

$$\lambda_{MPF\ Latent} = \lambda_{total} \times \left(1 - \frac{DC_{MPF\ Latent}}{100}\right)$$

Source: ASQF, Peter Lascych, 2011

Example Pedal Sensor Value



- As we do not have the overall failure rate yet, we have to use the EEC method

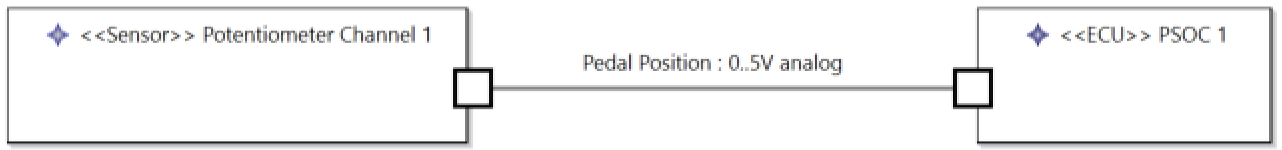
- Collect the faults
- Check if they effect the safety goal
- Check if the can be diagnosed
- Estimate the probability

Acc. table 1: KFD2-PT2-Ex1-* Potentiometer converter 1oo1 structure

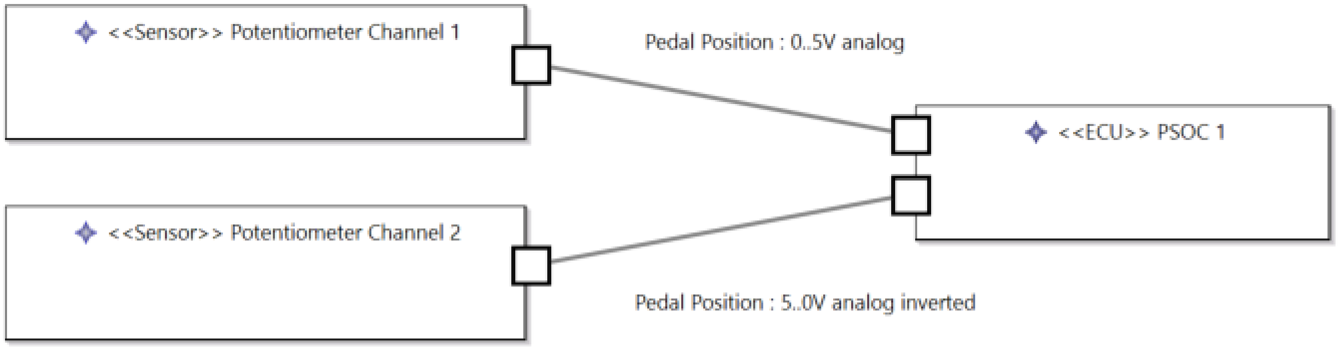
Parameters acc. to IEC61508	**Variables**
Device type	A
Demand mode	Low demand mode or High demand mode
Safety Function	Potentiometer converter
HFT	0
SIL	2
λ_{s}	0 FIT
λ_{dd}	0 FIT
λ_{du}	86.1 FIT
$\lambda_{\text{fail high}}$ [1]	6.35 FIT
$\lambda_{\text{fail low}}$ [1]	86.5 FIT
$\lambda_{\text{no effect}}$	218 FIT
$\lambda_{\text{total (Safety function)}}$	397 FIT
SFF	78.31%
MTBF[2]	265 years
PFH	$8.62*10^{-8}$ 1/h
PFD_{avg} for T_1 = 1 year	$3.77*10^{-4}$
$Tproof_{max}$	2.5 years

[1] "Fail high" and "fail low" failures are considered as dangerous detected failures.
[2] acc. To SN29500. This value includes failures which are not part of the safety function / MTTR = 8h

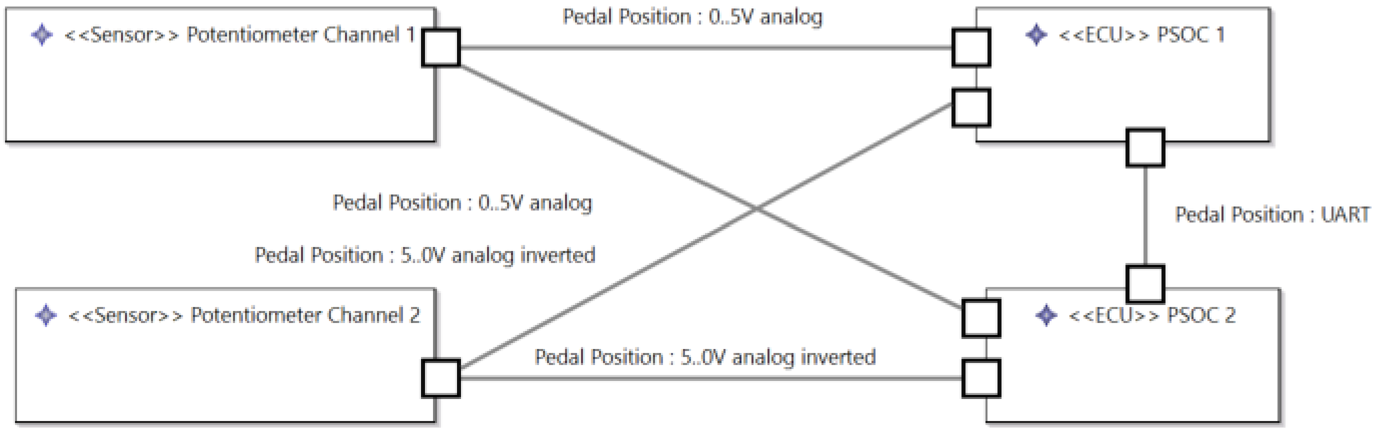
Design Alternatives



Option 1



Option 2



Option 3

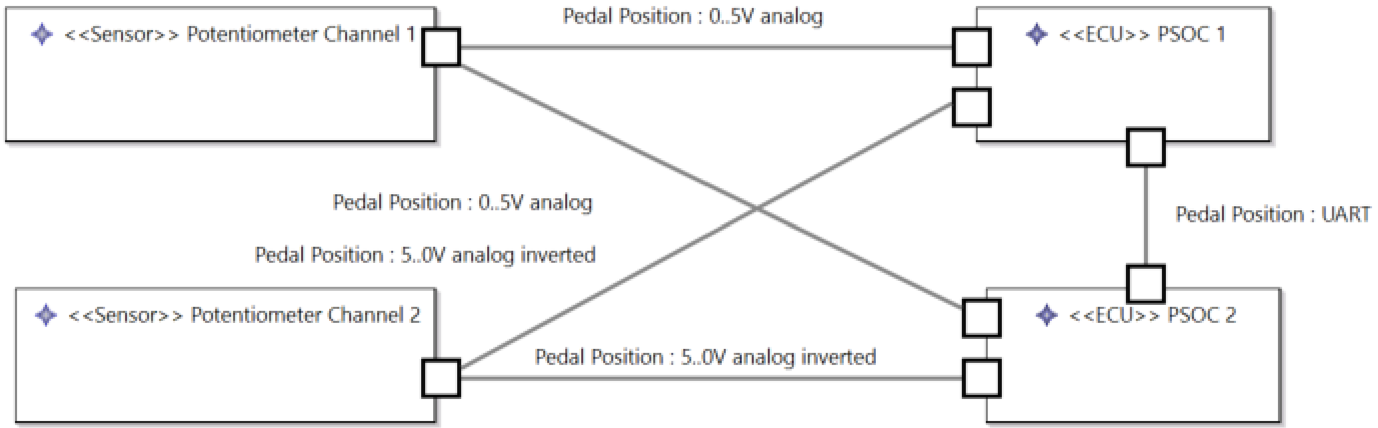
Single Point Fault analysis (qualitative, without considering failure rate of ADC)

Fault	Impact SG 1	Probability	DC with resistors	DC with redundancy
Potentiometer GND broken	Yes	10%	Yes	Yes
Potentiometer Vdd broken	Yes	10%	Yes	Yes
Potentiometer Signal broken	Yes	20%	Yes	Yes
Potentiometer Signal short to Gnd	Yes (No if value 0 implies safe state)	10%	Yes	Yes
Potentiometer Signal short to Vdd	Yes	10%	Yes	Yes
Potentiometer Signal bias	Yes	40%	No	Yes
Resulting DC			60% < 90%	>99,9%
Acceptable Failure Rate class			2 $< 10^{-9}$/h	5 $< 10^{-6}$/h (397 FIT ok)

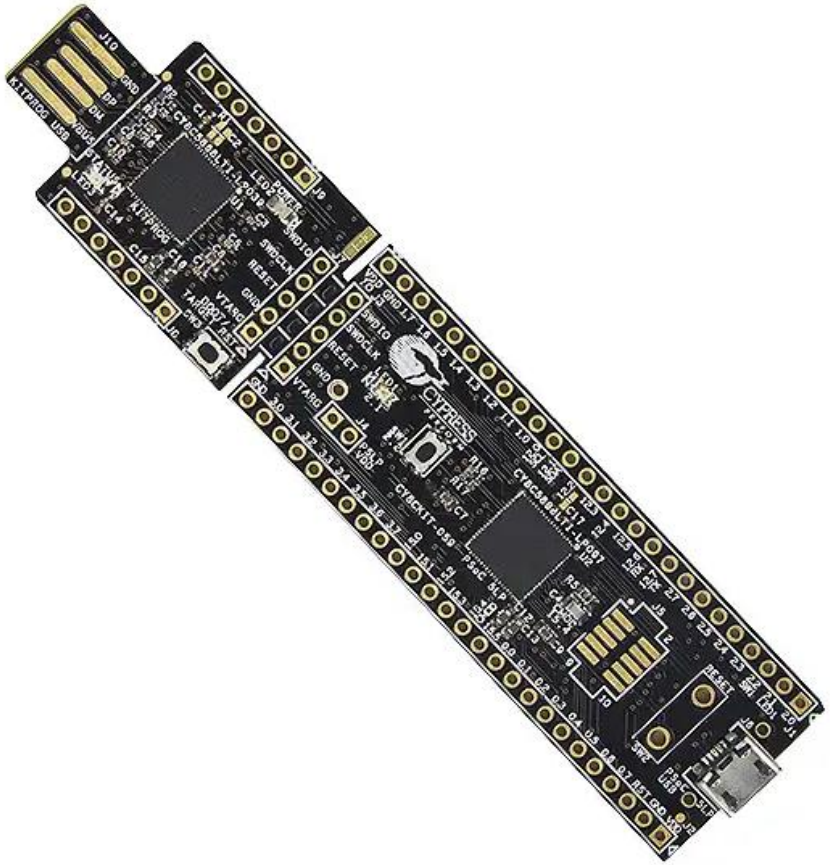
Redundancy CCF analysis (qualitative)

Potential CCF issues

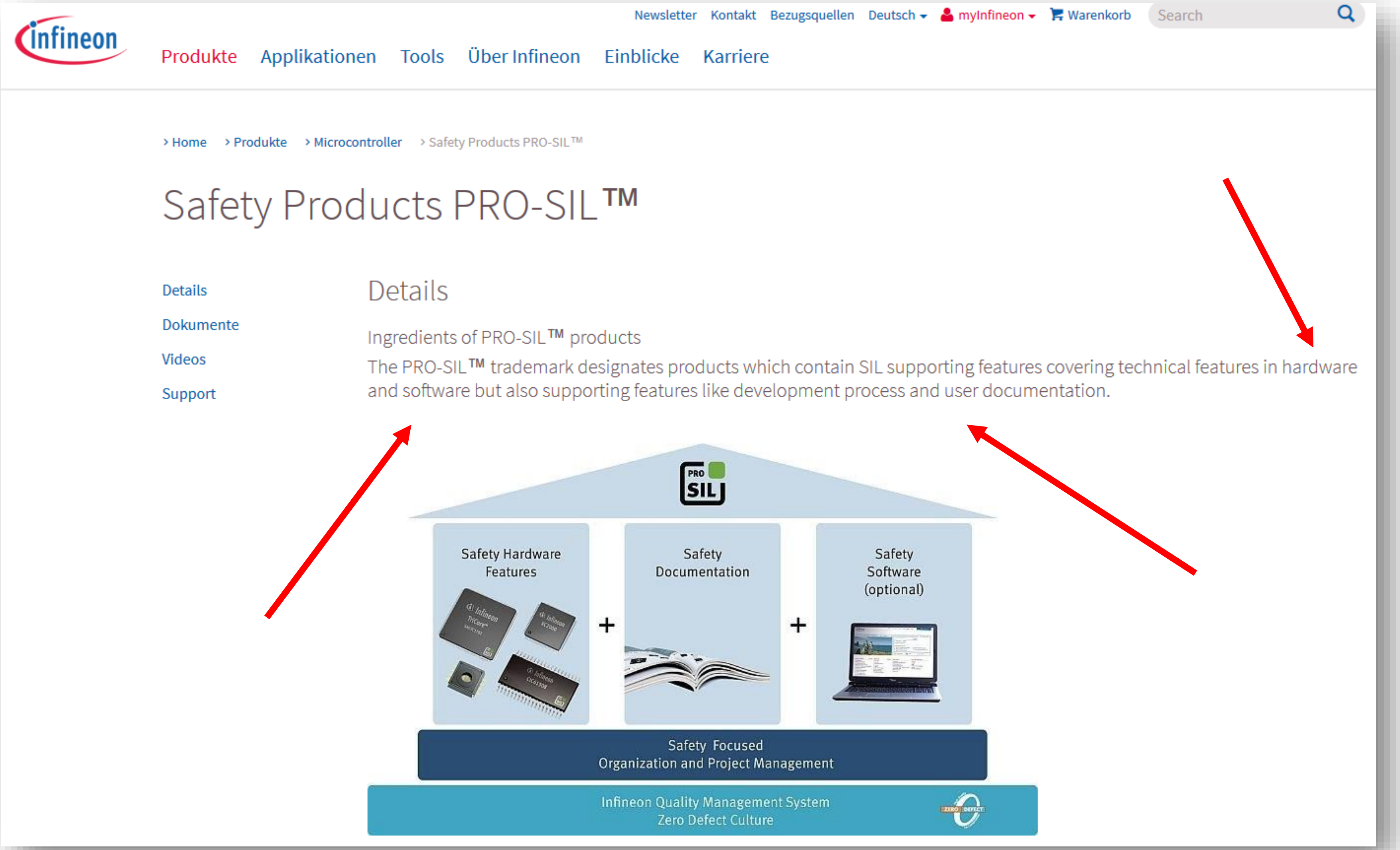
- Power loss
- ADC wrong configuration
- Microcontroller systematic peripheral fault
- Systematic software error



Safe microcontroller



Let’s check some marketing …



Safety Controller Features

- Memory Check (ECC)
- Logic supervision (Lockstep, multicore)

- Memory protection (MPU, MMU)
- Peripheral Protection (MPU, RAP)
- Timing control (Watchdog)

- Hardware error handling
  (Safety management Unit,
  Safe Pin, System Control Units)

Soft errors

Software errors

Escalation / Handling

Aurix Safety Controller

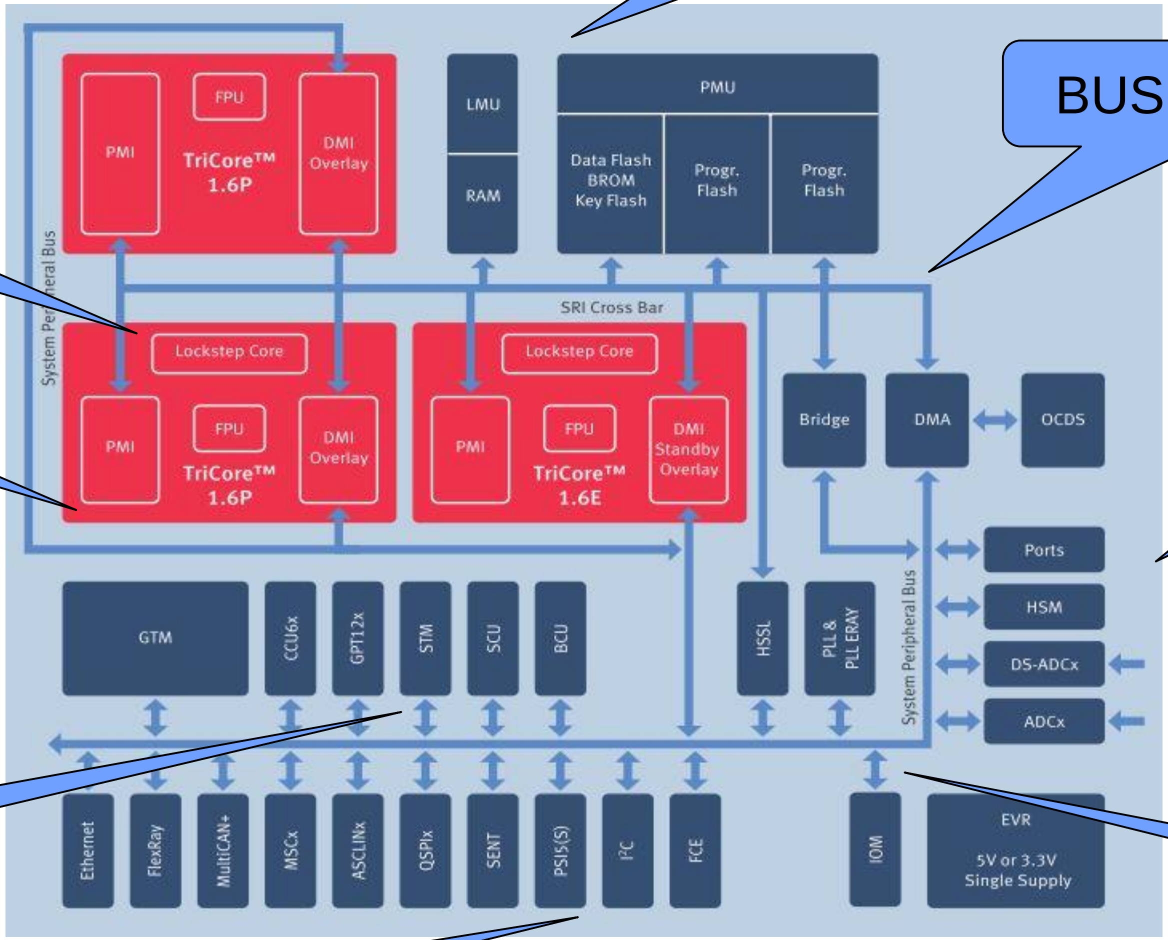
ECC

BUS Protection

Lockstep

MPU

RAP



SMU / SCU

Safe Pins

Internal and External Watchdogs

Soft Error

In electronics and computing, a soft error is a type of error where a **signal or datum is wrong**. Errors may be caused by a defect, usually understood either to be a **mistake in design or construction**, or a broken component….. One cause of soft errors is single event upsets from **cosmic rays**.

In a computer's memory system, a soft error changes an instruction in a program or a data value. …. A soft error will not damage a system's hardware; the only damage is to the data that is being processed.

There are two types of soft errors, **chip-level soft error and system-level soft error**. Chip-level soft errors occur when particles hit the chip, e.g., when secondary particles from cosmic rays land on the silicon die. If a **particle with certain properties hits a memory cell** it can cause the cell to change state to a different value. The atomic reaction in this example is so tiny that it does not damage the physical structure of the chip. System-level soft errors occur when the data being processed is hit with a **noise phenomenon, typically when the data is on a data bus**. The computer tries to interpret the noise as a data bit, which can cause errors in addressing or processing program code. The bad data bit can even be saved in memory and cause problems at a later time. (Source:Wikipedia)

If a soft error modifies data or logical behaviour related to a safety function

Challenge of Risk Estimation – Example Bit Flips

The error bitrate is somewhere between 1 failure per bit and hour up to 1 failure per bit and several years, depending on the cited source (data retrived 2020):

	Rate	Unit	per Memory	Scale to GB	Per 1GB and hour	MTTF [h]	MTTF [d]	MTTF[y]
SN29500	10	FIT	MB	1000	1,00E-05	100000	4166,667	11,41553
Wikipedia	1,00E-01	FIT	bit	8000000000	8,00E-01	1,25	0,052083	0,000143
Wikipedia	1,00E-08	FIT	bit	8000000000	8,00E-08	12500000	520833,3	1426,941
Toronto	25000	FIT	Mbit	8000	2,00E-01	5	0,208333	0,000571
Toronto	70000	FIT	MBit	8000	5,60E-01	1,785714286	0,074405	0,000204

1 3 2

Example small controller: STM G0, G4

1 2 3

- Up to 128kB RAM ➔ MTTF (200 days … 4,5 years … million of years)

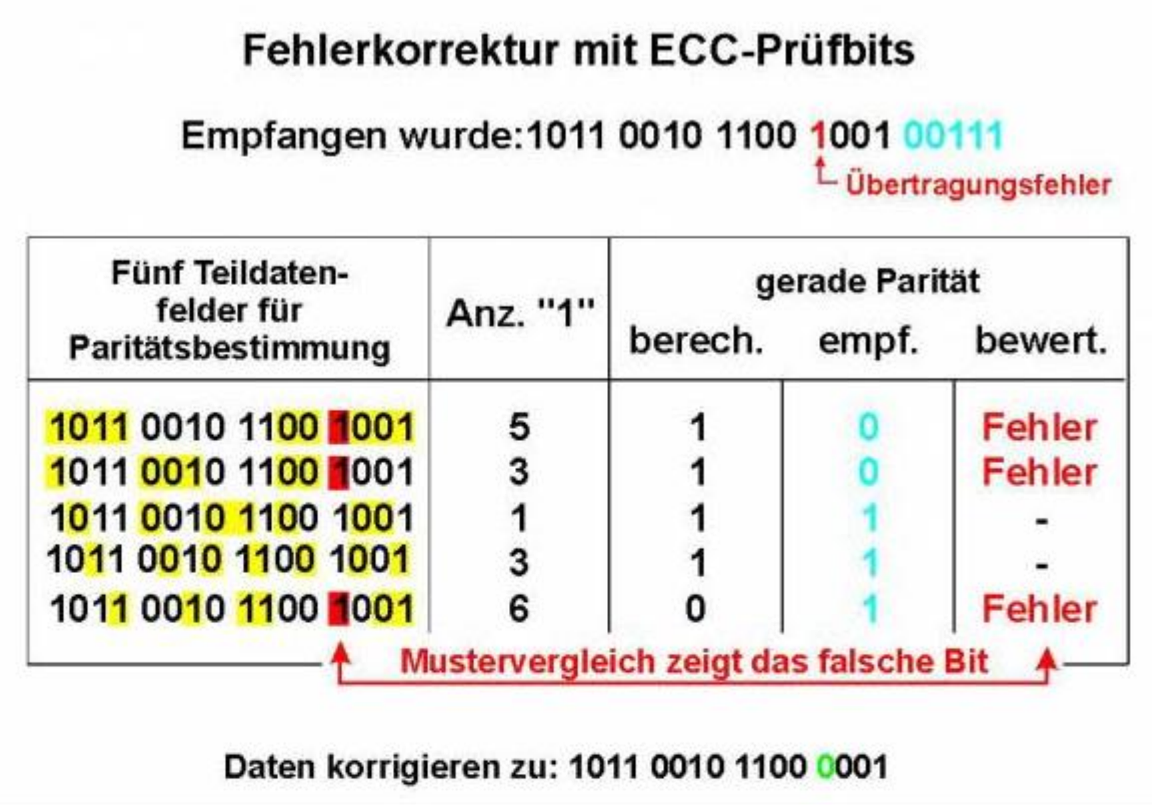
i.MX 8

- Up to 4GB RAM ➔ MTTF (20min … 1,25 hours … 3,5 years)

ECC

Error correction code memory (ECC memory) is a type of computer data storage that uses an error correction code (ECC) to detect and correct n-bit data corruption which occurs in memory. ECC memory is used in most computers where data corruption cannot be tolerated, like industrial control applications, critical databases, and infrastructural memory caches.

Typically, ECC memory maintains a memory system immune to single-bit errors: the data that is read from each word is always the same as the data that had been written to it, even if one of the bits actually stored has been flipped to the wrong state. Most non-ECC memory cannot detect errors, although some non-ECC memory with parity support allows detection but not correction. (Source: Wikipedia)



Fehlerkorrektur mit ECC-Prüfbits

Empfangen wurde:1011 0010 1100 1001 00111

Übertragungsfehler

Fünf Teildaten-felder für Paritätsbestimmung	Anz. "1"	gerade Parität berech.	empf.	bewert.
1011 0010 1100 1001	5	1	0	Fehler
1011 0010 1100 1001	3	1	0	Fehler
1011 0010 1100 1001	1	1	1	-
1011 0010 1100 1001	3	1	1	-
1011 0010 1100 1001	6	0	1	Fehler

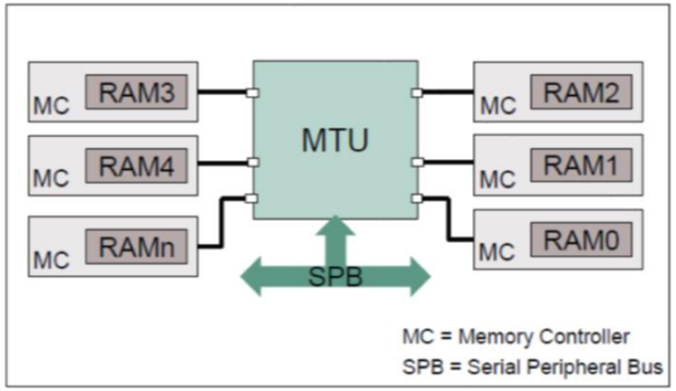
Mustervergleich zeigt das falsche Bit

Daten korrigieren zu: 1011 0010 1100 0001

Example Memory Test Unit Aurix (Infineon)

MTU
Memory Test Unit

infineon



Highlights

- The Memory Test Unit (MTU) integrates a unified interface for the control of Error Correction (ECC), Built-in-Self-Test (BIST) and redundancy features on the internal SRAMs within the AURIX™ family of microcontrollers.

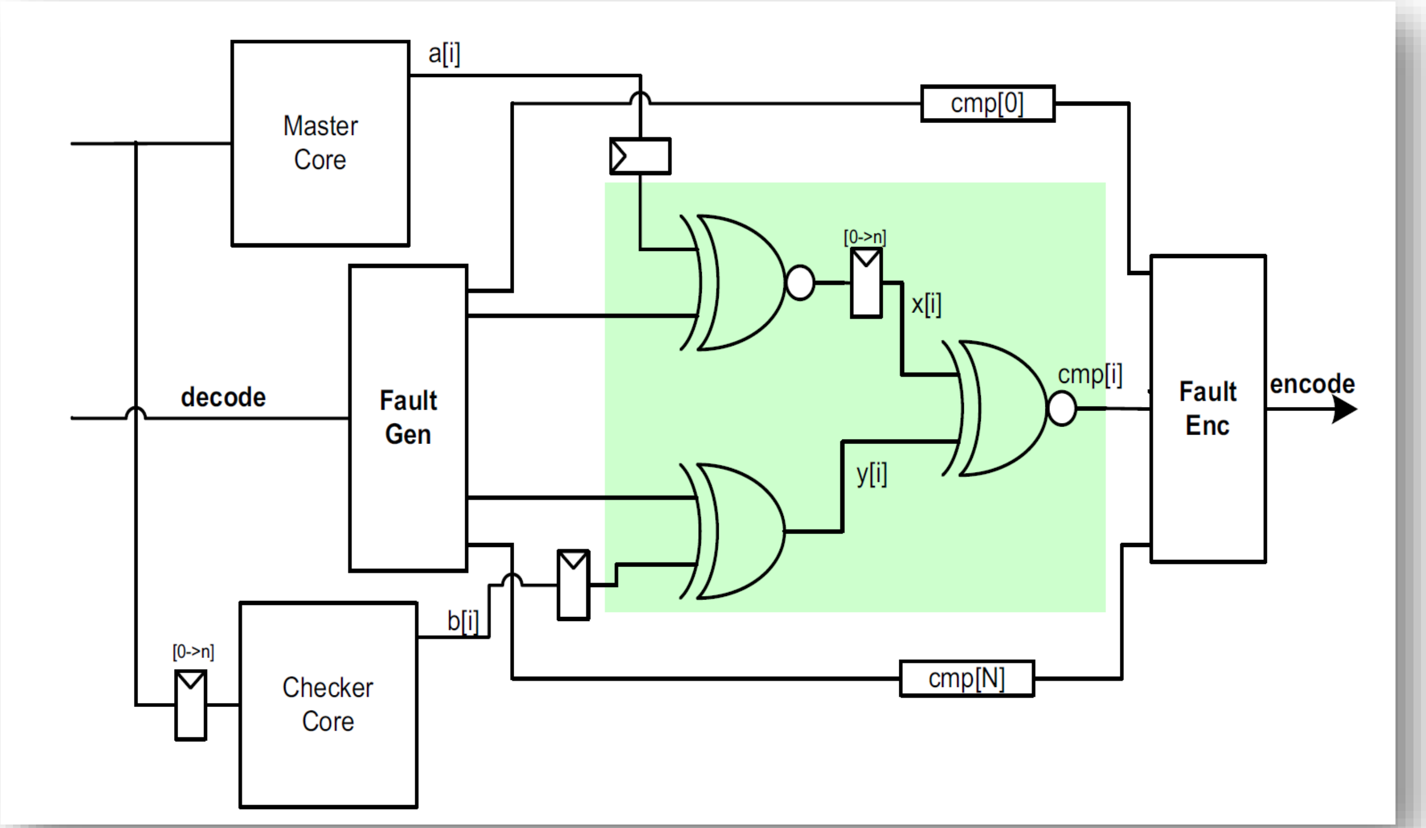
Key Features	Customer Benefits
Memory Initialization	› Fill the memories with a pre-defined data value for initialization
Memory Built-In-Self-Test (MBIST)	› Verification of integrity of the internal SRAMs realized in hardware

2019-03-27 2

For high (A)SIL levels, we must prove the correct operation of the hardware diagnostics. Special hardware error injection and test functions required!

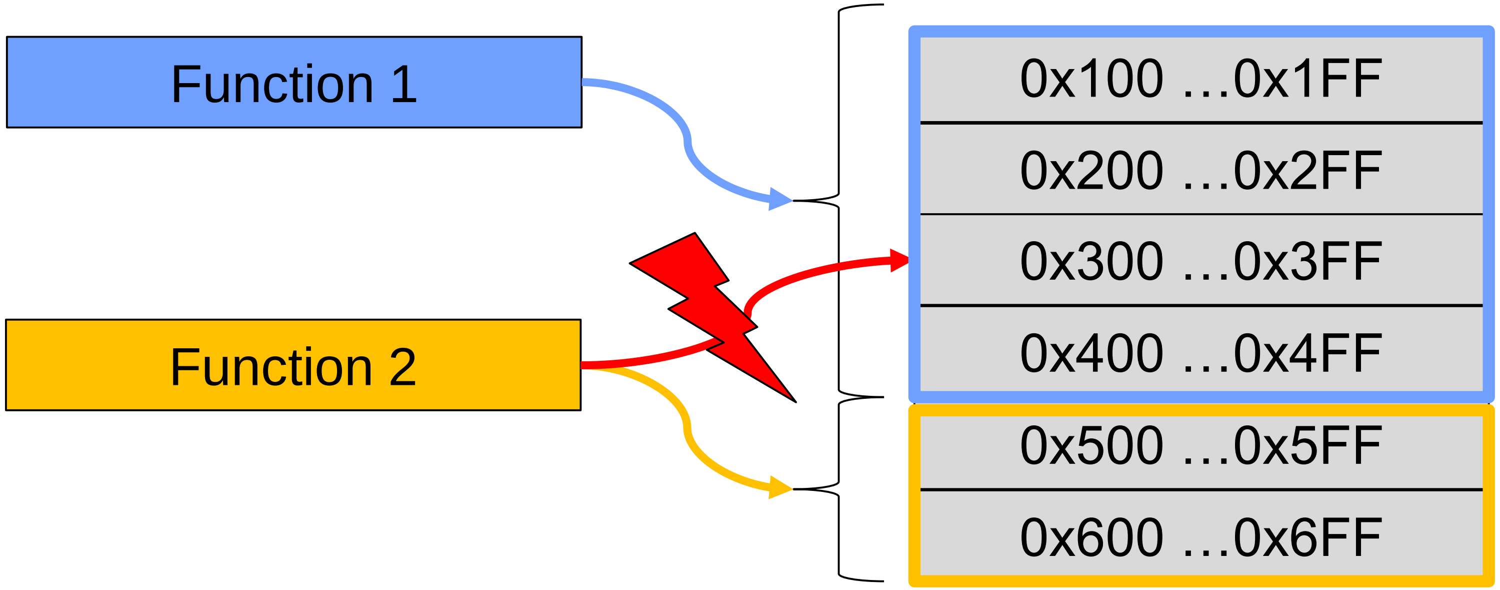
Lockstep

- Parallel execution of the code
- Comparison of results and error escalation in hardware



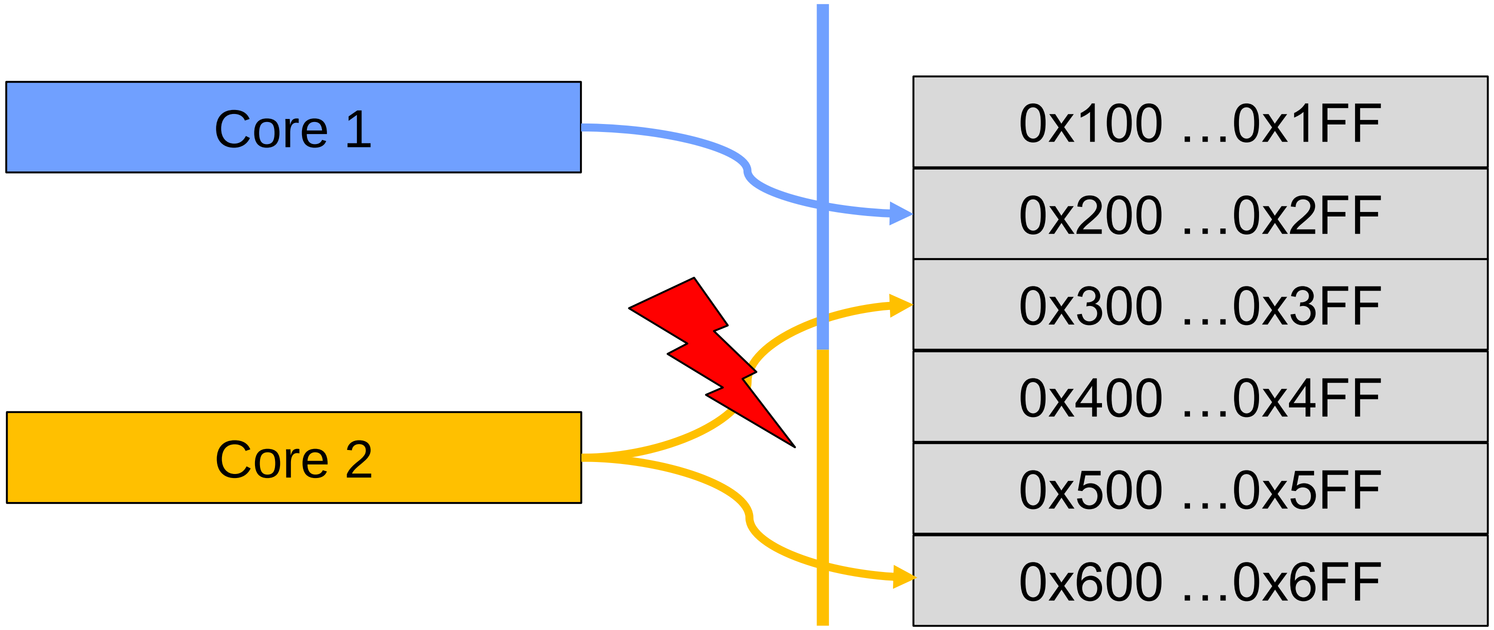
Memory Protection

- To ensure freedom from interference between software components, the ressources used by the components must be partitioned.
- For memory partitioning, a MPU is required.
- Different access modes (read / write / execute) are supported.

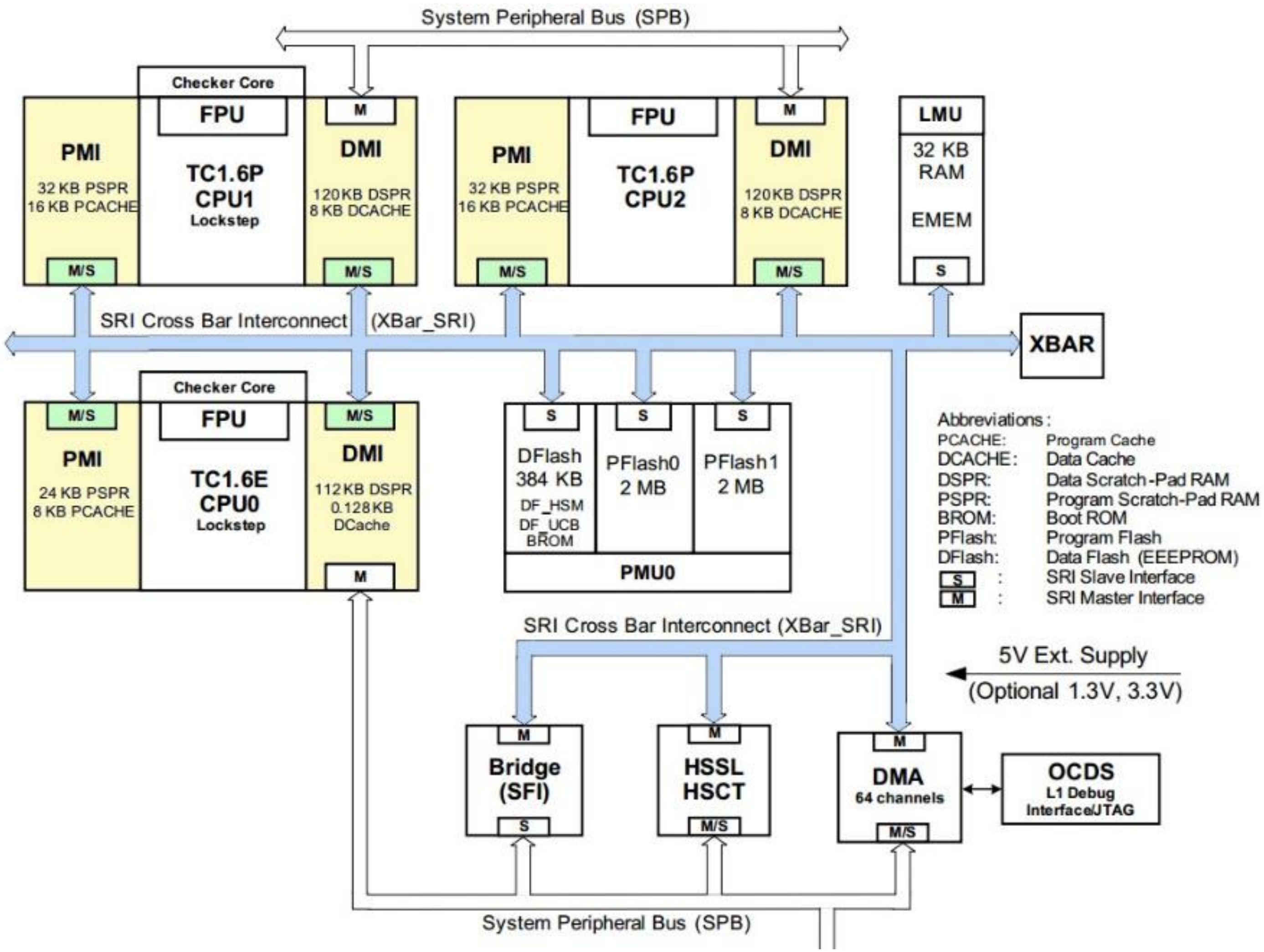


Bus protection

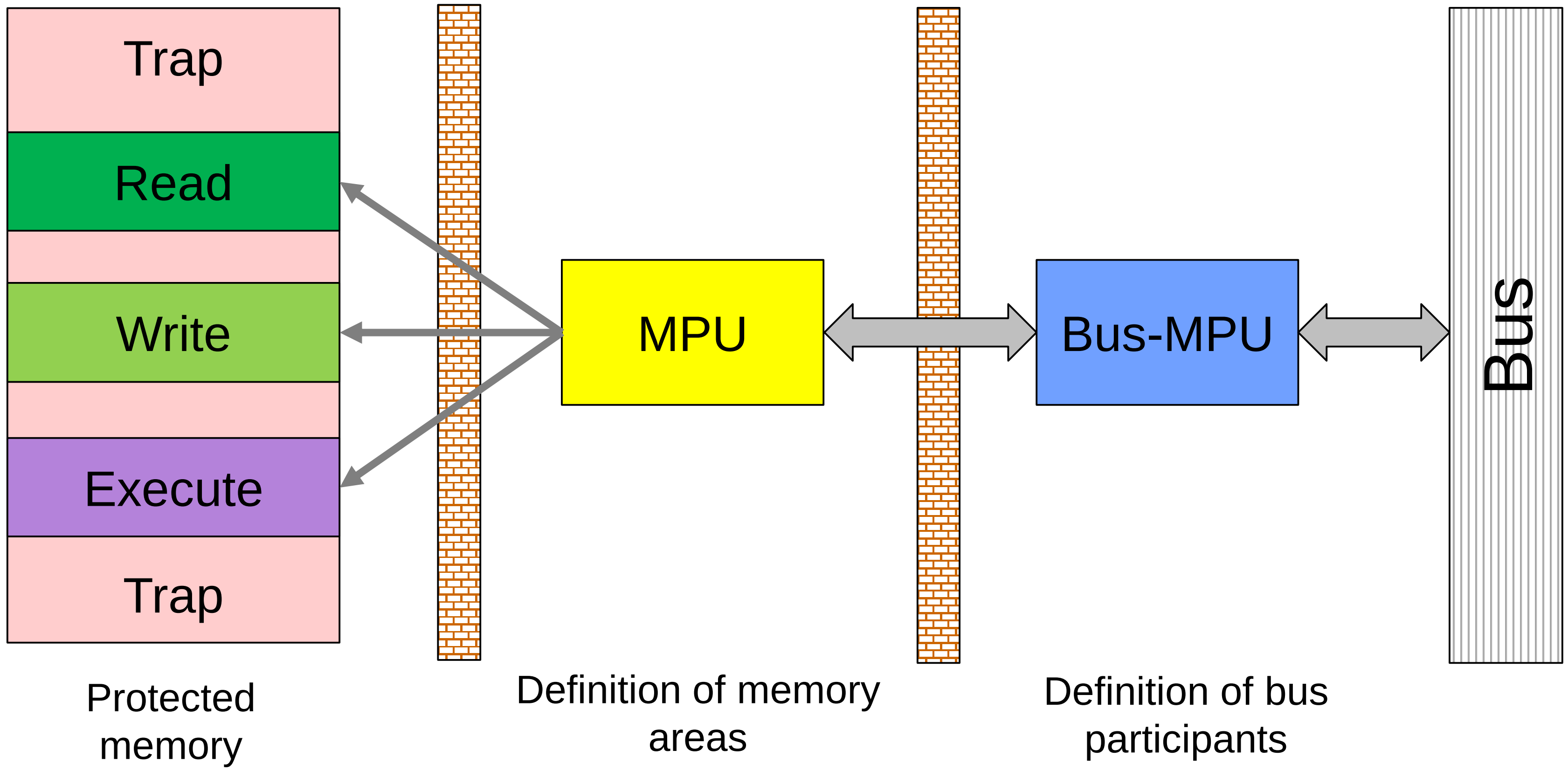
- Using a MPU, the application defines which areas can be accessed.
- Primary goal: protect against wild pointers.
- An additional protection is the so-called bus protection.
- Here, the access right of the CPU are limited by the bus. I.e. the peripheral unit defines, which core can access it.



Example Aurix Bus MPU



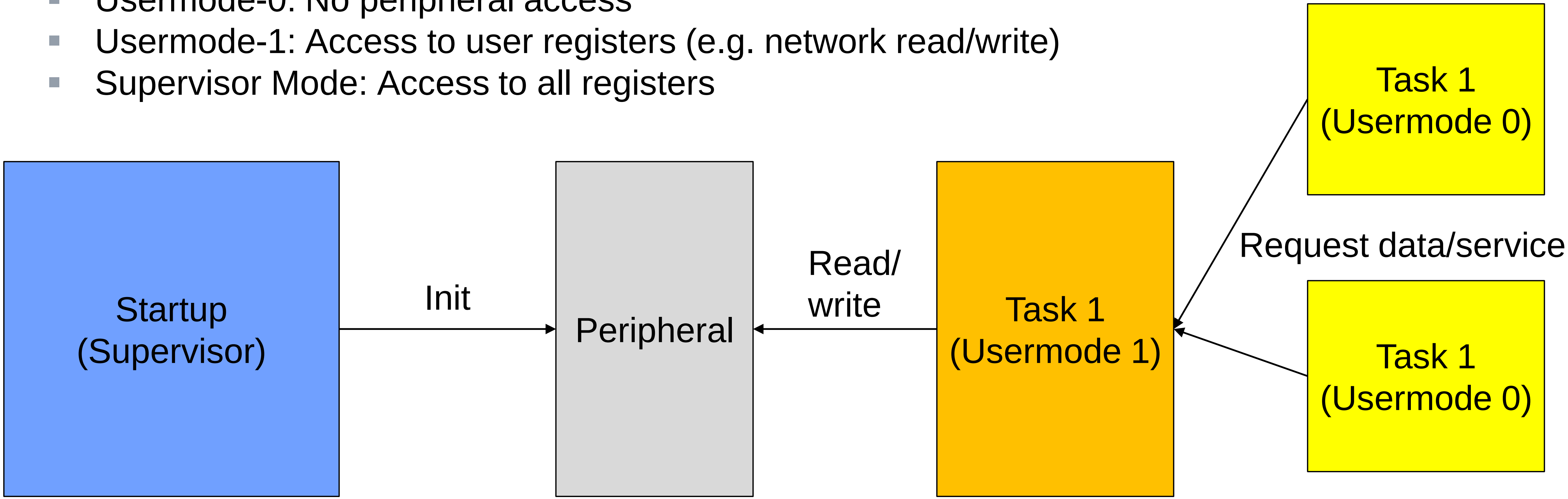
2 Layer “Firewall”



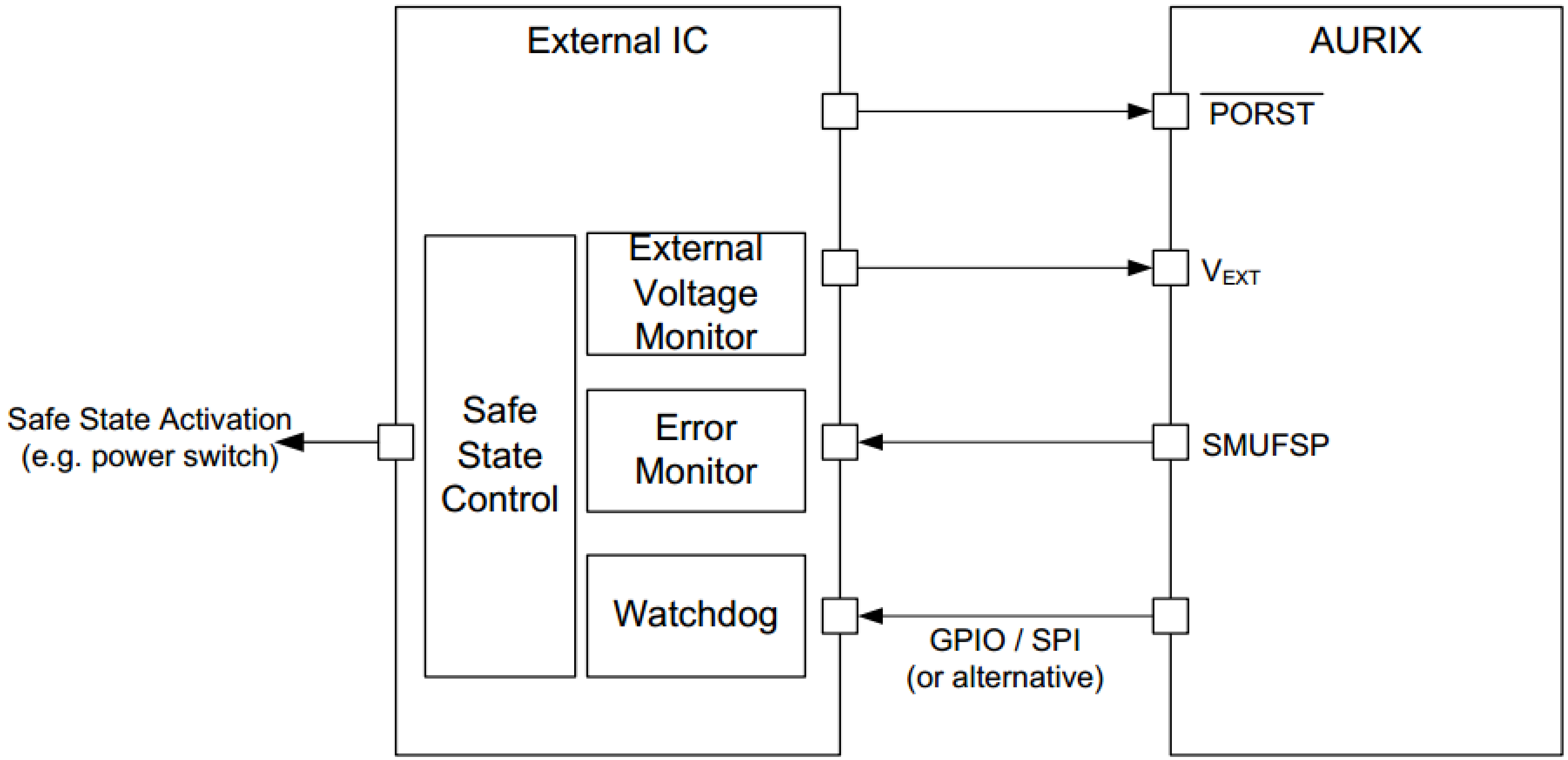
I/O Privilege Level

Depending on the user mode, different rights are provided. Typically handled by OS is cooperation with hardware

- Usermode-0: No peripheral access
- Usermode-1: Access to user registers (e.g. network read/write)
- Supervisor Mode: Access to all registers

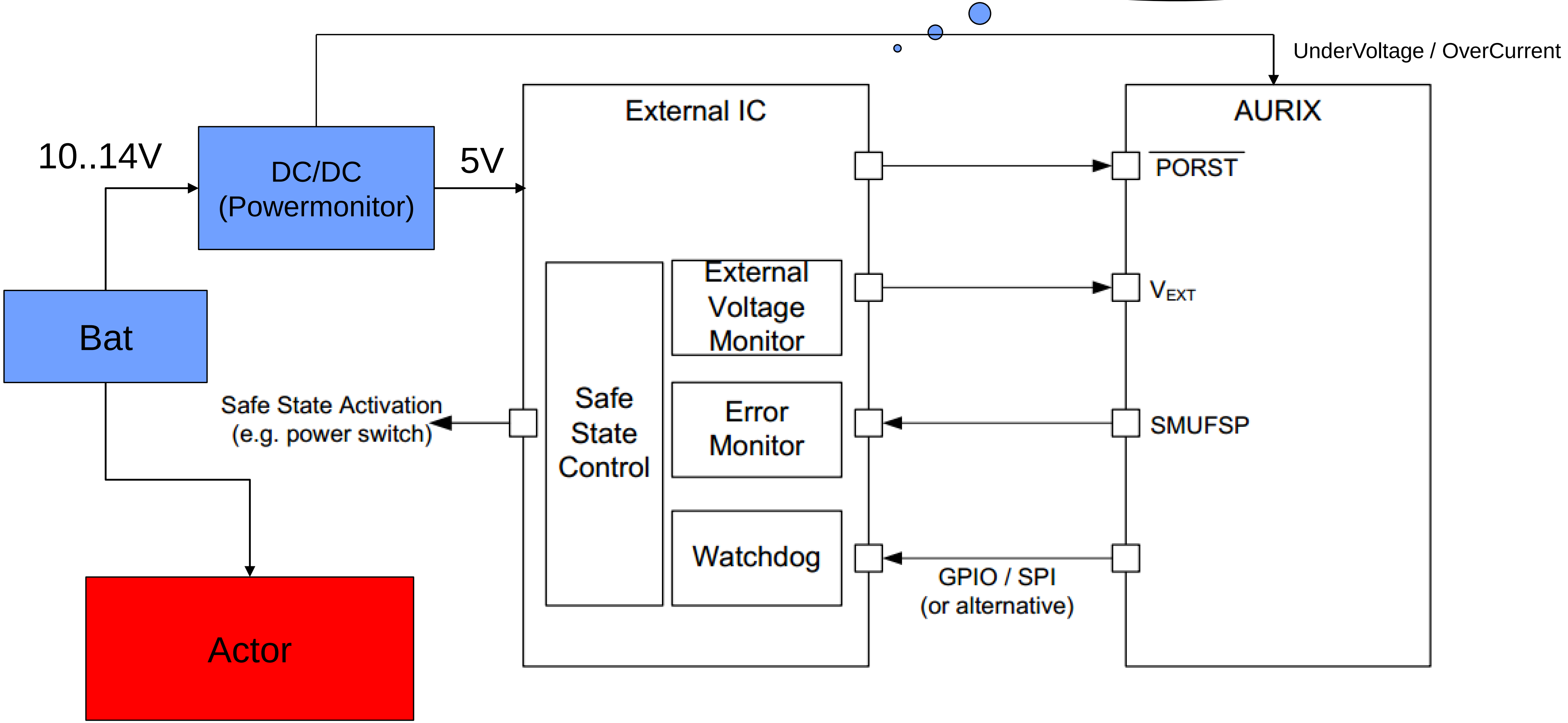


TLF Companion Chip

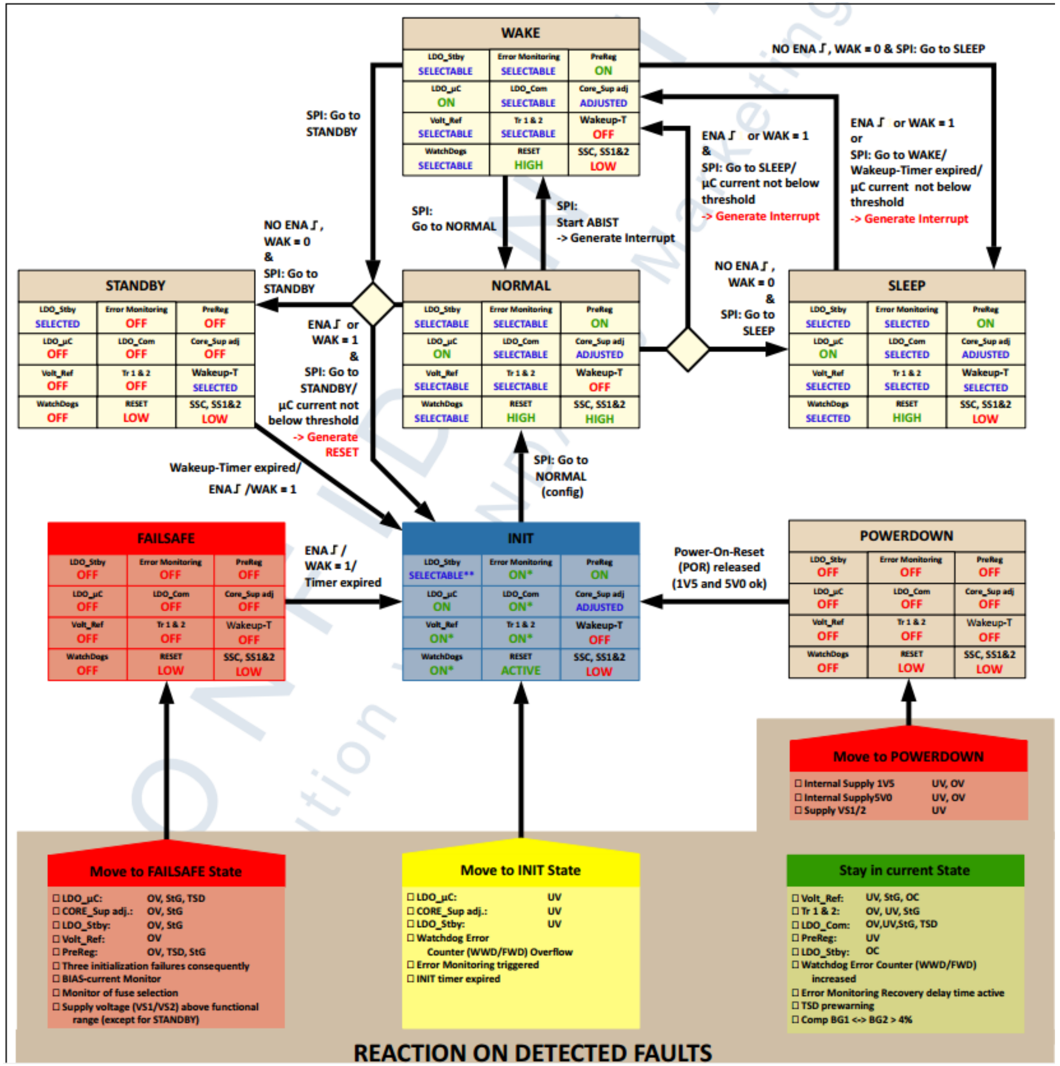


Challenge of Power Supervision

MCU (5V)still might be of in case of voltage drop, but the Actor not

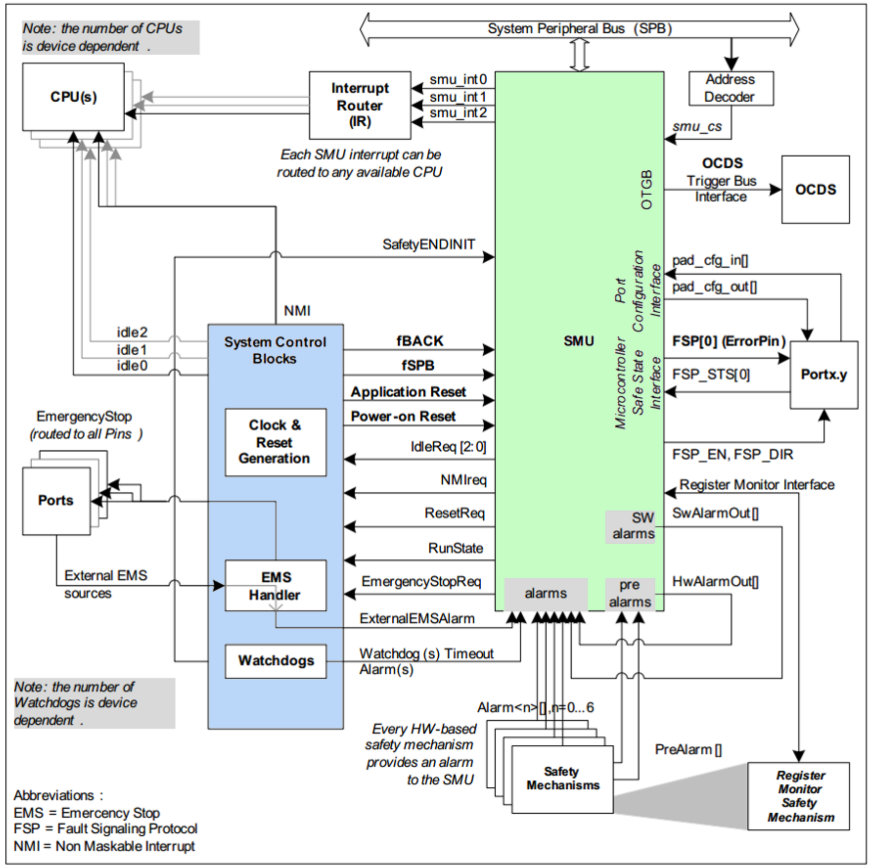


TLF Safety State Machine



Safety Management Unit

- Detection of hardware faults
- Configurable error handling (reset, turn of core, external escalation…)
- Safe pin handling



Comparison Safety MCU versus "normal" MCU

- Several features for hardware supervision

- Advanced access rights to protect memory and peripheral
- Timing supervision

- Detection of hardware faults
- Several 100 hardware registers need to be configured!

There is no alternative. We have to do this!

Safety Manual

AURIX Sa

Confidential

Table of Contents

4.5 Non-Lockstep CPU

Permanent random hardware faults such as register stuck-at and transient random hardware faults such as soft error caused by a neutron or alpha particle may cause an incorrect output of the CPU. The safety mechanisms listed in this section mitigate these errors.

4.5.1 Monitoring Concept

The permanent random hardware faults are random hardware faults in a non-lockstep C such as redundant software execution.

Attention: [SM_AURIX_NLSCPU_4] *The software lockstep CPU reaches 90% di model. For safety goals with shall be used.* [/req]

Attention: [SM_AURIX_NLSCPU_5] *The hardware Self-Test, SM1[AoU].CPU:SBST, with regard to permanent failu the influence of transient failure*

4.25.1.1 Redundant Acquisition Using Two VADC Channels

In this scenario, illustrated in Figure 8, two analog signals are converted into digital information by using two independent VADC channels.

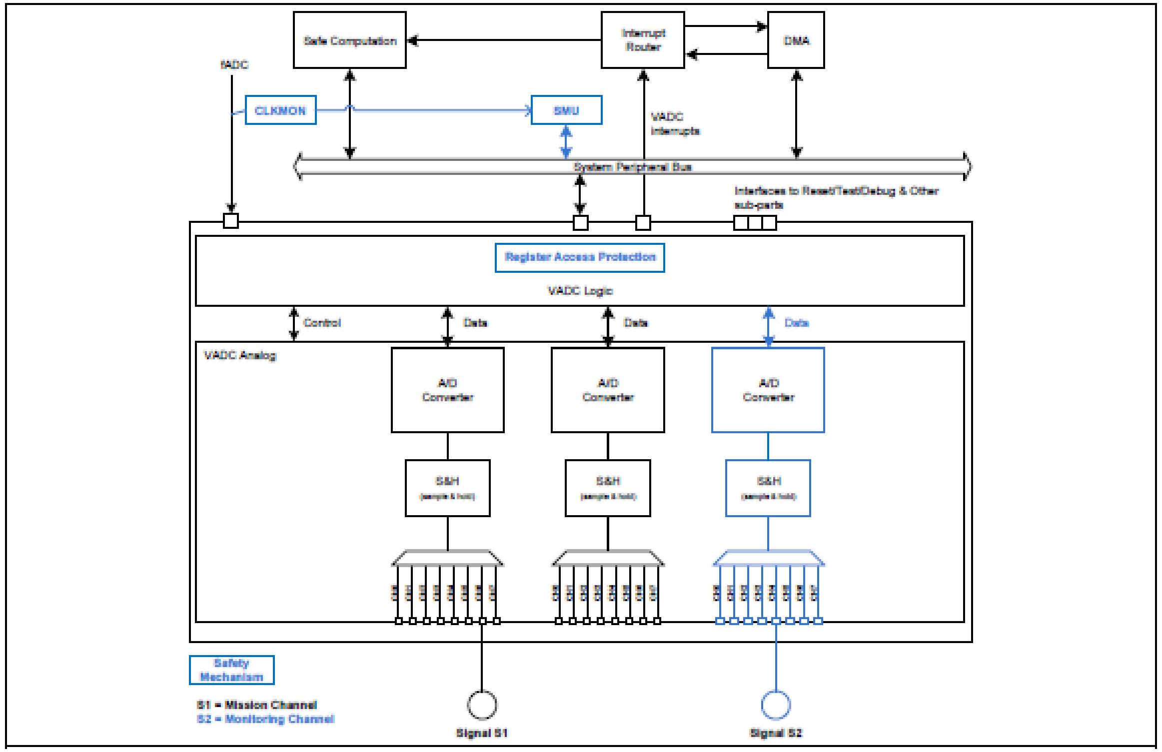
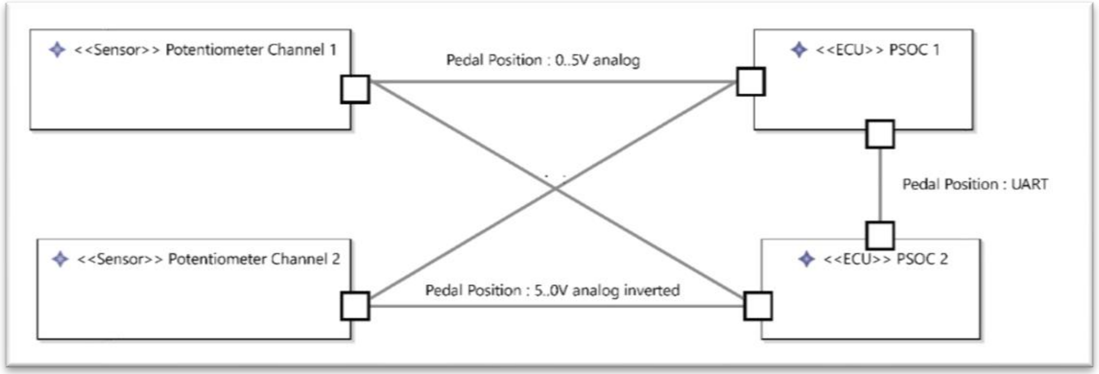


Figure 7 Redundant Acquisition Using Two VADC Channels

The mystery of pins

Mystery of Pins 1
Sensor signal comparison – USART

Let's consider the following setup:



- Both sensors are received by both MCU
- In addition PSOC1 transmits the value of Sensor1 via UART TX to PSOC2
- PSOC2 transmits the value of Sensor2

- What happens if PSOC 1 or 2 is without power?

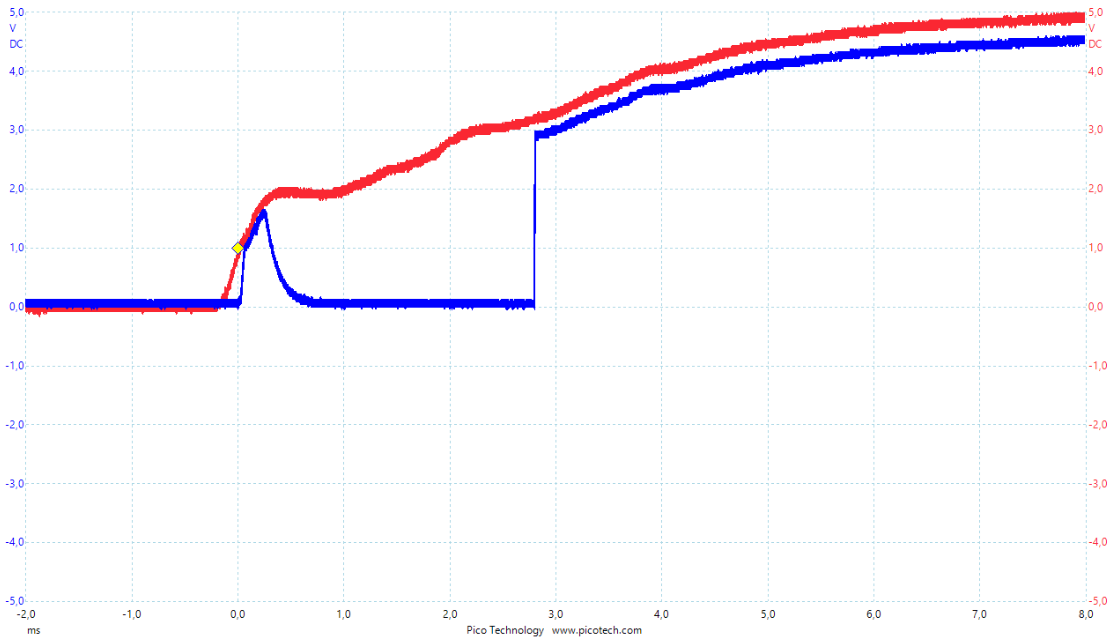
Expected: No data or zero or crap is being transmitted.

Reality: It depends on the pins. You might receive an echo (cross-talk) of the transmitting PSOC. I.e. it looks like everything is perfectly ok.

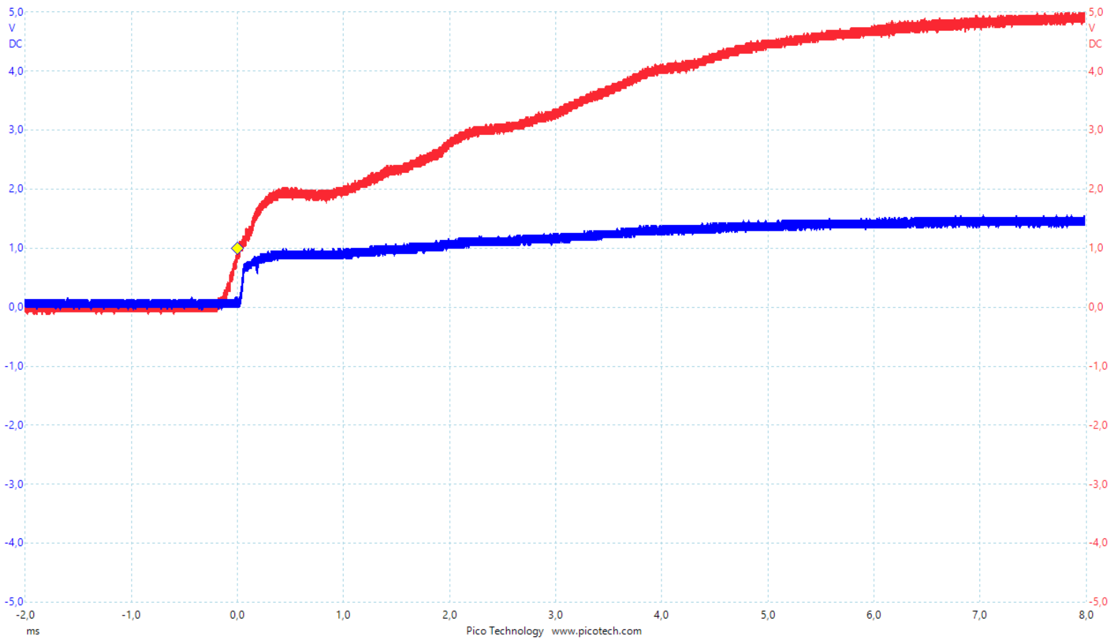
Mystery of Pins 2
Power on Behavior of Output Pin

- The watchdog PSOC drives a transistor to control the relay of the engine.
- The default value is 0, but the software set’s the pin high upon startup.

Expected Power-On



Reality Power-On



Red: Power supply
Blue: Pin

Mystery of Pins 2
Power on Behavior of Output Pin

We did not pull the bastard down????

- Somewhere, on page 500ff of the user manual…
- In all the devices in the PSoC 4000 family, P1[6] is temporarily configured as the XRES pin during power-up for approximately 100 ms, until the device enters the start-up code. Care should be taken when using this pin. P1[6] SHOULD NOT BE PULLED DOWN DURING POWER-UP because the device will enter and remain in the XRES state after power-up, as the reset signal is always asserted."

10	P1.6/OVF0/UND0/nOUT0 /CMPO_0	18	P1.6/OVF0/UND0/nOUT0 /CMPO_0	12	P1.6/OVF0/UND0/nOUT0/CMPO_0	14	P1.6/OVF0/UND0/nOUT0/CMPO_0	7	P1.6/OVF0/UND0/nOUT0/CMPO_0	nOUT0: Complement of OUT0, UND0, OVF0 as above	CMPO_0: Sense Comp Out, Internal Reset function[1]

Note
1. Must not have load to ground during POR (should be an output).

Wrapup

Focus of the hardware design is the reliability of components

- Mandatory for ASIL B - D
- ASIL A does not require hardware certification
- Moving from A to B or higher with existing solutions is difficult / not possible. Hardware redesign required!

Architectural level

- Redundancy (hardware fault tolerance)
- Diagnostic Coverage of possible faults (single point, latent, multiple fault, residual)

Safety Controller

- Hardware safety features (hardware diagnostics)
- Resource protection (MPU, watchdog,…)
- Safety Manual (guidelines)
- Interface to external protection units